OCCUPATIONAL HEALTH, SAFETY & ENVIRONMENT

# JAFURAH FIELD GAS PLANT

## OPERATIONAL FACILITY / EXPANSION INTERFACE RISK ASSESSMENT

CONTROLLED DOCUMENT | ACTIVITY-BASED HAZARD AND RISK REGISTER | PUBLIC-SOURCE PRE-ASSESSMENT

**DOCUMENT STATUS: NOT APPROVED FOR EXECUTION — SITE DATA, WORK PACK AND OWNER REQUIREMENTS MUST BE VERIFIED**

Public-source pre-assessment — site validation and authorized approval required. This document follows a generic professional reference format for layout guidance; it is NOT an official Aramco/client form, is NOT Aramco-approved, and is not evidence of a site inspection.

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## 1. DOCUMENT CONTROL & PROJECT INFORMATION

*Complete every applicable field with verified information. Use "Not applicable" only where justified; do not invent missing site information.*

| | | | |
|---|---|---|---|
| **Project / Facility**<br>Saudi Aramco Jafurah Gas Development — Jafurah Field Gas Plant, Eastern Province, Saudi Arabia | **Client / Asset Owner**<br>Saudi Aramco (operator / developer) [S-01][S-02]. Midstream: Jafurah Midstream Gas Company holds development/usage rights under a lease-and-leaseback; Aramco retains 51% [S-03] | **Principal Contractor**<br>[SITE INPUT REQUIRED — EPC/main contractor for the selected package is not publicly confirmed] | **Subcontractor(s)**<br>[SITE INPUT REQUIRED — not publicly disclosed] |
| **Assessment Title / Activity**<br>Operational facility / expansion interface risk assessment — construction, tie-in, pre-commissioning and commissioning activities interfacing with the live gas plant, where confirmed in the approved work pack | **Work Location / Area**<br>Jafurah Field Gas Plant [CONFIRM exact unit / plot with the approved plot plan] | **RA Number / Document ID**<br>JAF-RA-001 [TO BE CONFIRMED BY DOCUMENT CONTROL] | **Revision / Status**<br>Rev. 0 — Public-source pre-assessment — NOT APPROVED FOR EXECUTION |
| **Assessment Date**<br>28 October 2026 (public-source research snapshot) | **Review / Expiry Date**<br>[PER PROJECT PROCEDURE — to be set at approval; review triggers are listed in Section 15] | **Related PTW / JSA / Method Statement**<br>[SITE INPUT REQUIRED — verified references to be entered from the site PTW register and the approved work pack] | **Applicable Procedure / Standard**<br>Owner HSE / PTW procedures [NOT REVIEWED — access-controlled, S-10]; ISO 45001:2018 + Amd 1:2024 (framework) [S-07]; Saudi HRSD midday-work decision [S-06] |
| **Prepared By / Role**<br>[REAL NAME / TITLE / SIGNATURE — to be completed by an authorized competent person] | **Reviewed By / Role**<br>[REAL NAMES / TITLES — Operations, Construction, Process/Mechanical/Electrical, HSE] | **Approver / Role**<br>[AUTHORIZED APPROVER ONLY — this Rev. 0 pre-assessment is NOT approved] | **Workforce / Shifts**<br>Current site headcount NOT PUBLICLY DISCLOSED — obtain gate / access-control and contractor mobilisation records (Section 5) |

### 1.1 Document control detail

| Document control item | Entry |
|---|---|
| **Document number** | JAF-RA-001 [TO BE CONFIRMED BY DOCUMENT CONTROL] |
| **Project / contract number** | [CONFIRM — to be assigned from the actual work package] |
| **Facility / area** | Jafurah Field Gas Plant, Eastern Province, Saudi Arabia — exact plot/area [CONFIRM WITH APPROVED PLOT PLAN] |
| **Assessment boundary** | Operating gas-processing facility and adjacent expansion / construction / commissioning interfaces at the Jafurah location only. EXCLUDES: Riyas NGL Fractionation Facility (Jubail) and the Master Gas System Phase III programme — separate sites/programmes. |

| Document control item | Entry |
|---|---|
| **Work package / work order** | [CONFIRM — assessment applies only to activities confirmed in the approved work pack] |
| **Revision** | Rev. 0 — Public-source pre-assessment |
| **Assessment (issue) date** | 28 October 2026 (public-source research snapshot) |
| **Date of site walkdown / work-pack review** | [TO BE COMPLETED — no site visit has been made for this pre-assessment] |
| **Prepared by** | [REAL NAME / TITLE / SIGNATURE — to be completed by an authorized competent person] |
| **Reviewed by** | [REAL NAMES / TITLES / SIGNATURES — Operations, Construction, Process/Mechanical/Electrical, HSE] |
| **Approved by** | [REAL NAME / TITLE / SIGNATURE — authorized approver only. This document is NOT approved.] |
| **Classification** | Working pre-assessment — not an official Aramco document; not Aramco-approved; contains no Aramco branding and claims no review of access-controlled Aramco manuals. |

*VALIDATION DISCLAIMER: This assessment was prepared from publicly available sources only (see Section 17 references, access date 28 October 2026). No site visit, worker interview, permit review, drawing review or headcount verification has taken place. Every site-specific value that could not be verified from public sources is marked [CONFIRM] / [SITE INPUT REQUIRED] and must be established from controlled site documents and a competent site walkdown before this document is approved or used to authorize work.*

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## 2. Executive summary

Why this assessment exists. Aramco's Jafurah unconventional gas development in Saudi Arabia's Eastern Province has started production (December 2025, [S-01]) while major expansion work — gas compression facilities and associated pipelines, expansion of the Jafurah Gas Plant with additional gas-processing trains, utilities, sulfur and export facilities — was contracted under Phase Two awards announced on 30 June 2024 ([S-02]). Construction, tie-in, pre-commissioning and commissioning activity therefore interfaces with a live, pressurised natural-gas processing facility. That operating/expansion interface is the highest-risk feature of the project and is the focus of this assessment.

Top major-accident hazards identified (Section 7): loss of containment of flammable gas/condensate with jet fire, flash fire or vapour cloud explosion (RA-03); line opening on incompletely isolated or unproven systems (RA-10); confined-space entry (RA-13); possible toxic-gas (H2S) exposure, which is a VERIFICATION-CRITICAL hazard because gas composition is not public (RA-04); failure of isolation/LOTO (RA-09); heavy lifting over or near live plant (RA-17/RA-18); and breakdown of the emergency-response interface for a large transient construction workforce (RA-41). Heat stress is a critical, legally regulated hazard in the Saudi summer (RA-37, [S-06]).

Critical controls and hold points. Work must not start until, as a minimum: the SIMOPS interface is reviewed and authorised by operations; isolations are proven with zero-energy verification; gas composition data has been received and the H2S decision recorded; the owner risk matrix has replaced the provisional matrix in Section 6; the approved emergency plan details are briefed to every worker; and actual shift headcount is being captured for muster reconciliation. Each hold point is stated on the relevant register row and in Section 8.

Unresolved inputs. The current total headcount inside the Jafurah boundary is NOT publicly disclosed and is not estimated anywhere in this document. Process conditions, hazardous-area classification, exact workfronts and the current work package are likewise unconfirmed. Section 14 lists every site input required before competent review and approval. Residual risk is therefore NOT declared acceptable by this pre-assessment: residual ratings are provisional professional judgements that only become valid when the stated controls are implemented, verified and accepted by the authorized owner representative.

## 3. Project profile and source-backed facts

Labels used throughout: 'Verified public fact' = checked against the cited source on 28 October 2026; 'Historically reported' = true for its stated period/scope only; 'Technical consideration — not yet confirmed at site' = a credible hazard or control topic that site data must confirm or rule out; 'Site input required' = information that only controlled site documents can supply.

| Fact | Value / statement | Status & source | What it does — and does not — prove |
|---|---|---|---|
| Operator / project | Saudi Aramco — Jafurah unconventional gas development / Jafurah Field Gas Plant, Eastern Province, Saudi Arabia. | Verified public fact [S-01][S-02] | Establishes the real project and operator. Does NOT establish the assessment's exact plot boundary — plot plan required. |
| Production status | Production started December 2025. | Verified public fact [S-01] | Proves the facility is an operating plant, so construction interfaces with live hydrocarbon systems. Does NOT give today's production rate. |
| 2030 ambition (targets) | ~2.0 bscfd sales gas; 420 MMscfd ethane; ~630,000 bpd gas liquids and condensates by 2030. | Verified public fact — PLANNED TARGETS [S-01] | Shows intended scale/direction. These are future targets, NOT current measured rates, and must not be used as process design data. |
| Reserves (context) | 229 trillion scf raw gas; 75 billion STB condensate (field estimate). | Verified public fact [S-02] | Context for field scale only. No bearing on workfront-level hazards. |
| Phase Two scope (announced 30 Jun 2024) | 16 contracts, ~US$12.4bn: gas compression facilities + associated pipelines; Jafurah Gas Plant expansion incl. gas-processing trains; utilities; sulfur and export facilities. | Verified public fact [S-02] | Defines the announced scope elements this RA addresses. Does NOT prove every element is active in today's work package. |
| Separate facility — EXCLUDED | Riyas NGL Fractionation Facility is in JUBAIL, a different location, although part of the wider Jafurah development and the same contract announcement. | Verified public fact [S-02] | Riyas is outside this assessment's boundary. Mixing the two sites would corrupt headcount, emergency and SIMOPS data. |
| Separate programme — EXCLUDED | Master Gas System Phase III: ~4,000 km of pipelines and 17 new gas compression trains across the Kingdom (~US$8.8bn). | Verified public fact [S-02] | A Kingdom-wide transmission programme, not one Jafurah site. Excluded from this RA. |
| Midstream arrangement | US$11bn lease-and-leaseback (announced 14 Aug 2025): | Verified public fact [S-03] | Ownership/financing context. Day-to-day site control and permit authority |

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| Fact | Value / statement | Status & source | What it does — and does not — prove |
|---|---|---|---|
| | Jafurah Midstream Gas Company holds development/usage rights for the Jafurah Field Gas Plant and Riyas facility, leased back to Aramco for 20 years; Aramco retains 51% and the exclusive right to receive, process and treat Jafurah raw gas. | | must still be confirmed from the site organisation — NOT inferred from this announcement. |
| Current contractor activity | A Philippine government report (5 June 2026) identifies an electrical & instrument supervisor employed by Hyundai Engineering & Construction assigned to the Jafurah Gas Plant project. | Verified public fact [S-08] | Corroborates ongoing E&I/construction activity at the project in 2026. Does NOT evidence the contractor list, package split or headcount. |
| Owner safety manuals | Aramco supplier resources include Contractor Workforce Qualification Assurance Program materials that are password-protected for registered contractors. | Verified public fact [S-10] | Proves owner construction/safety requirements exist in controlled form — and that they were NOT reviewed for this pre-assessment. They must be obtained through authorized channels. |
| Heat regulation | HRSD 2026 ban on work under direct sunlight: 15 June – 15 September 2026, 12:00–15:00, private-sector establishments; heat-prevention duties continue year-round. | Verified public fact — SAUDI REGULATORY REQUIREMENT [S-06] | Sets the legal floor for RA-37 for the 2026 season. If work falls in a later season, that season's notice must be verified. |
| Management-system framework | ISO 45001:2018, confirmed in 2024, remains current; Amendment 1:2024 (climate action) applies; a revision is in progress (stage 90.92 at access date). | Verified public fact [S-07] | Provides the OH&S management framework (hazard identification, worker participation, emergency planning). It is NOT a hazard checklist and NOT Saudi law. |

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# 4. Site footprint, asset and workfront register

Public sources describe the facility elements below at project level. No public source provides a current approved workfront register, package split or plot boundary; every operational field is therefore [SITE INPUT REQUIRED]. Riyas (Jubail) is listed only to record its exclusion.

| Facility / asset / workfront | Location | Package / contractor | Current activity & SIMOPS interface | Operational status | Source / drawing reference | Data owner | Verification status |
|---|---|---|---|---|---|---|---|
| Jafurah Field Gas Plant — existing operating trains | Eastern Province [CONFIRM plot/unit IDs] | [CONFIRM package] | Live operation; construction interface where tie-ins/modifications occur | Operating | Approved plot plan [SITE INPUT REQUIRED] | Operations | Not verified — site input required |
| Gas-processing trains (Phase Two expansion) | Jafurah [CONFIRM] | [CONFIRM] | Construction / installation / commissioning adjacent to live plant | Announced scope [S-02]; current activity [CONFIRM] | Phase Two announcement [S-02]; work pack [SITE INPUT REQUIRED] | Construction Manager | Scope verified publicly; activity not verified |
| Gas compression facilities + associated pipelines | Jafurah [CONFIRM] | [CONFIRM] | Mechanical/piping installation; pipeline tie-in interfaces | Announced scope [S-02]; current activity [CONFIRM] | Announcement [S-02] | Construction Manager | Scope verified publicly; activity not verified |
| Utilities, sulfur and export facilities (within Jafurah package only) | Jafurah [CONFIRM] | [CONFIRM] | Installation/commissioning interface; sulfur-area work ONLY if confirmed in package and process data | Announced scope [S-02]; current activity [CONFIRM] | Announcement [S-02] | Construction Manager | Scope verified publicly; sulfur hazards NOT confirmed — see RA register verification notes |
| Well tie-in / gathering interfaces | Jafurah field area [CONFIRM] | [CONFIRM] | Tie-in work only where the approved work pack confirms it | 13 well tie-in contracts were awarded Dec 2022–May 2024 [S-02]; current status [CONFIRM] | Announcement [S-02] | Operations | Not verified — do not assume drilling activity (excluded) |
| Fabrication / laydown / scaffold yard areas | [CONFIRM locations] | [CONFIRM] | Prefabrication, storage, module handling; traffic interface | A contractor yard ~1 km from plot was reported historically [S-05]; current arrangement [CONFIRM] | KAEFER story [S-05] (historical) | Construction Manager | Historical only — confirm current arrangement |
| Riyas NGL Fractionation Facility — JUBAIL | Jubail — SEPARATE SITE | n/a | NONE — excluded from this assessment | Operating/ development [S-02][S-03] | Announcements [S-02][S-03] | n/a | EXCLUDED — separate assessment required if work is performed there |
| Master Gas System Phase III network | Kingdom-wide — SEPARATE PROGRAMME | n/a | NONE — excluded from this assessment | Programme in progress [S-02] | Announcement [S-02] | n/a | EXCLUDED — not a Jafurah site |

# 5. Workforce and exposure register

## 5.1 Public figures — with date, scope and correct use

| Figure | Value | Date / scope | Correct use in this assessment |
|---|---|---|---|
| Saudi Aramco company employees | 76,664 | As of 31 December 2025 — whole company (Aramco 'at a glance') [S-04] | Corporate context ONLY. Not a Jafurah headcount; excludes contractor/subcontractor personnel who perform most construction work. |
| Saudi construction-sector registered | 3,348,143 | Q2 2026 — national sector data via | National sector context ONLY. Not Aramco's workforce and not Jafurah site headcount. |

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| Figure | Value | Date / scope | Correct use in this assessment |
|---|---|---|---|
| workers | | DataSaudi/GASTAT [S-09, not re-verified] | |
| KAEFER Jafurah project crew (one contractor, Phase 1 scope) | More than 400 on site in 2024; around 700 planned for the following year | KAEFER's own project story, 2024 [S-05] | HISTORICAL, contractor-specific manpower for KAEFER's access/coating/fireproofing/refractory/insulation scope only. The 700 figure was an expectation, not a confirmed actual. Neither figure is a total site workforce or a 2026 figure. |
| Total persons physically inside the Jafurah boundary on the issue date | NOT PUBLICLY DISCLOSED | Current — not found in any public source reviewed | MUST be obtained from the gate/access-control report and contractor mobilisation/daily manpower reports. Do not estimate. |
| Number of active workfronts / packages on the issue date | NOT CONFIRMED IN PUBLIC SOURCES | Current | Public announcements describe packages let in 2024 [S-02]; the live workfront register must come from site management. |

**WARNING: Aramco's corporate employee total, the national construction-sector total and one contractor's historical crew size measure three different things. None of them is the number of people inside the Jafurah fence today, and none may be used for muster, welfare, transport or emergency planning.**

## 5.2 Actual manpower and exposure register — TO BE COMPLETED FROM SITE RECORDS

*Source of count required: gate access-control / badge report, muster roster, contractor mobilisation register or daily manpower report. Until a dated record is attached, every field below remains 'Not publicly disclosed — obtain actual site record.'*

| Field | Value (from site record only) | Record reference |
|---|---|---|
| Headcount date and exact time of count | Not publicly disclosed — obtain actual site record | |
| Shift pattern in force (day / night / both) [CONFIRM] | Not publicly disclosed — obtain actual site record | |
| Direct operator (owner) personnel inside boundary — day shift | Not publicly disclosed — obtain actual site record | |
| Direct operator personnel — night shift | Not publicly disclosed — obtain actual site record | |
| EPC / main contractor personnel — day shift | Not publicly disclosed — obtain actual site record | |
| EPC / main contractor personnel — night shift | Not publicly disclosed — obtain actual site record | |
| Subcontractor personnel (list each subcontractor separately) — day | Not publicly disclosed — obtain actual site record | |
| Subcontractor personnel — night | Not publicly disclosed — obtain actual site record | |
| Vendor / OEM / commissioning personnel | Not publicly disclosed — obtain actual site record | |
| Visitors inside boundary | Not publicly disclosed — obtain actual site record | |
| TOTAL persons inside the defined boundary (this count) | Not publicly disclosed — obtain actual site record | |
| Peak concurrent workforce in the reporting period | Not publicly disclosed — obtain actual site record | |
| Personnel working at height (this shift) | Not publicly disclosed — obtain actual site record | |
| Confined-space entrants and attendants (this shift) | Not publicly disclosed — obtain actual site record | |
| Lifting crew (operators, riggers, signallers) | Not publicly disclosed — obtain actual site record | |
| Hot-work crew and fire watches | Not publicly disclosed — obtain actual site record | |
| Electrical / instrument crew | Not publicly disclosed — obtain actual site record | |
| Plant / traffic interface personnel | Not publicly disclosed — obtain actual site record | |
| Emergency-response personnel on shift (fire, medical, rescue) | Not publicly disclosed — obtain actual site record | |
| Prepared by (name/role) / Verified by (name/role) / Revision & reporting period | Not publicly disclosed — obtain actual site record | |

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# 6. Assessment methodology and applicable requirements

Method. Work is decomposed into activity steps (Section 7). For each step the assessor identifies the hazard source and the specific hazardous event, the persons exposed and how, and the credible consequences — including fatality and major-accident consequences where they are foreseeable. Controls already in place are recorded separately from additional controls, and each existing control is marked with its evidence status: in this pre-assessment almost all are 'to be verified', because no site evidence has been reviewed. Additional controls follow the hierarchy of controls (elimination, substitution, engineering, administrative, PPE), prioritising isolation and engineering barriers over procedures and PPE. A risk rating is never reduced merely by writing 'follow procedure' or adding PPE.

Three different risk values. (1) Initial/inherent risk assumes no controls beyond those physically inherent in the task. (2) Current risk credits only existing controls that are implemented AND verified by evidence. (3) Residual risk is rated only after the additional controls are implemented and independently/competently checked. In this draft, initial and residual ratings are PROVISIONAL PROFESSIONAL JUDGEMENTS made for comparison and prioritisation; they must be re-rated in the site validation workshop.

Major-accident limitation. A 5x5 matrix is a screening and prioritisation tool. It is NOT sufficient on its own for major-accident process hazards: a low likelihood score must never be used to dismiss a catastrophic scenario (RA-03, RA-04, RA-08). Such scenarios require the project's specialist studies (HAZID, HAZOP, LOPA, QRA, bow-tie) and the interface to those studies is a site input (Section 14). This document is an activity-based HSE risk assessment; it does not replace a process hazard analysis, a method statement, or task-level JSAs (Section 9).

## 6.1 Proposed 5 x 5 risk matrix — PROPOSED WORKING MATRIX ONLY

*Use the client/project-approved risk matrix whenever one exists. The 5 x 5 matrix below is a PROPOSED reference method only and must be replaced or formally accepted by the responsible organization before use. Risk score = Likelihood x Severity. The owner/client risk matrix was not available; likelihood/severity definitions are proposed qualitative definitions pending approval.*

### PROPOSED 5 x 5 RISK MATRIX

| S x L | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 5 | 5 | 10 | 15 | 20 | 25 |
| 4 | 4 | 8 | 12 | 16 | 20 |
| 3 | 3 | 6 | 9 | 12 | 15 |
| 2 | 2 | 4 | 6 | 8 | 10 |
| 1 | 1 | 2 | 3 | 4 | 5 |

### RATING DEFINITIONS (TO BE APPROVED)

#### LIKELIHOOD SCALE

| Level | Term | Definition / guide |
|---|---|---|
| 1 | Rare | Exceptional; highly unlikely in the defined conditions. |
| 2 | Unlikely | Not expected, but credible. |
| 3 | Possible | Could occur under some circumstances. |
| 4 | Likely | Expected to occur in many circumstances. |
| 5 | Almost certain | Expected to occur frequently / repeatedly. |

#### SEVERITY SCALE

| Level | Term | Definition / guide |
|---|---|---|
| 1 | Insignificant | No injury or negligible harm / damage. |
| 2 | Minor | First-aid case or minor, reversible damage. |
| 3 | Moderate | Medical-treatment / lost-time potential; significant reversible harm or damage. |
| 4 | Major | Single-fatality potential, life-changing injury or major damage. |
| 5 | Catastrophic | Multiple fatalities or catastrophic asset / environmental consequences. |

**PROPOSED SCORING BANDS (subject to project approval): 1–4 Low — manage by routine controls. 5–9 Moderate — specific controls and supervision. 10–16 High — additional controls, documented review and authorized acceptance required before work. 17–25 Extreme — work must not proceed until risk is reduced and acceptance is given at the level the client's system requires. These bands are proposed working bands, not a universal ISO/Aramco matrix. High/Extreme residual risks in Section 7 carry explicit hold points.**

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## 6.2 Applicable requirements and verification register

| Requirement / source | Jurisdiction & type | Relevant topic (summarised, not reproduced) | Applicability | Version/date checked | Owner/client document still needed | Verification status |
|---|---|---|---|---|---|---|
| Saudi HRSD decision — ban on work under direct sunlight [S-06] | Saudi Arabia — LAW / REGULATORY | No work under direct sunlight 12:00–15:00 during the seasonal ban period (2026: 15 Jun–15 Sep); heat-stress prevention duties continue year-round | All outdoor work in season | 2026 notice, concluded 15 Sep 2026; checked 28 October 2026 | Current-season notice for the actual work dates; site heat-stress procedure | Requirement verified; site procedure outstanding |
| Saudi labour / OSH framework (general) | Saudi Arabia — LAW | Employer duties for worker safety, hours, welfare — exact current texts to be confirmed by the project's legal/HSE function | All work | Not verified in this check | Project legal register | SITE INPUT REQUIRED — do not cite clause numbers from memory |
| ISO 45001:2018 + Amd 1:2024 [S-07] | International — MANAGEMENT SYSTEM STANDARD | Framework for hazard identification, worker participation, operational control, emergency preparedness and review | Assessment process framework | Confirmed current 2024; checked 28 October 2026 | Project HSE management plan mapping | Verified as framework only — not a legal requirement |
| Owner (Aramco) HSE / construction safety requirements [S-10] | Client-specific — CONTRACTUAL | PTW, isolation, lifting, confined space, gas testing, emergency and competency requirements (CWQAP) | All work — governing documents for execution | Access-controlled; NOT reviewed | Current authorized copies via the project's document control | NOT REVIEWED — critical site input |
| OSHA PSM overview & hazard recognition [S-13][S-14] | United States — TECHNICAL GUIDANCE ONLY | Process-safety management concepts for hydrocarbon interface work | Technical reference for RA-01/03/04/07/08/10 | Checked 28 October 2026 | Owner process-safety standards (governing) | Guidance only — NOT Saudi law |
| OSHA confined spaces; welding/cutting guidance [S-15][S-16] | United States — TECHNICAL GUIDANCE ONLY | Control concepts for confined-space entry, hot work and welding fume | Technical reference for RA-05/13/30/31 | As cited in dossier | Owner confined-space and hot-work procedures (governing) | Guidance only — NOT Saudi law |

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# 7. Detailed activity-based risk assessment register

*Use one entry for each meaningful work step / hazard scenario. Hazards are described specifically; existing controls are distinguished from additional controls; residual risk is rated only for controls that are implemented and verified. 43 specific entries. Ratings use the PROVISIONAL matrix in Section 6.1 and are professional judgements for prioritisation only — they must be re-rated in the site validation workshop with the owner matrix. 'Existing controls' in this pre-assessment are stated with their evidence status; a control without evidence is NOT credited in any residual rating until verified on site.*

| RA ID | Work activity / step | Asset / area / workfront | Hazard / hazardous event | Potential consequence & persons at risk | Existing controls | Initial risk (L x S = score) | Additional controls / actions (hierarchy of controls) | Responsible role & target date | Residual risk (L x S = score) | PTW / procedure reference | Verification evidence / status |
|---|---|---|---|---|---|---|---|---|---|---|---|
| RA-01 | Construction / installation work adjacent to operating gas-processing equipment | Jafurah Field Gas Plant — live process areas [CONFIRM unit/plot] | Impact, dropped load, vehicle strike or hot particle contact with live hydrocarbon equipment causing loss of containment (LOC). | Hydrocarbon release, fire/explosion, multiple fatalities, major plant damage and escalation. Persons at risk: Construction crews, plant operators and adjacent contractors within the affected area; exposure via physical proximity to live equipment. | TO BE VERIFIED: plant boundary/barricading, permit-to-work controls and operations oversight are owner systems — no site evidence reviewed for this pre-assessment. | 3 x 5 = 15 — HIGH | ELIMINATION: schedule intrusive/adjacent work during planned shutdown where feasible. ENGINEERING: physical barriers/impact protection between workfront and live equipment; defined equipment-protection zones on the plot plan. ADMIN: SIMOPS review and written operations authorisation before work starts; daily coordination meeting with Area Authority; exclusion of unauthorised personnel. PPE: task PPE per site standard (last line only). | Area Authority / Operations Representative Priority: Critical Target date: [SITE INPUT REQUIRED] | 2 x 5 = 10 — HIGH | Owner PTW (cold/hot work as applicable); SIMOPS authorisation; approved plot plan extract Review trigger / reference: Review on any change of plant status or workfront. Ref: OSHA PSM overview (technical guidance) [S-13]. | Evidence status: To be verified — no site evidence reviewed HOLD POINT: no work within the agreed protection zone of live equipment without signed operations authorisation. Stop work on any plant alarm/upset until Area Authority clears resumption. |
| RA-02 | Simultaneous operations (SIMOPS): hot work concurrent with hydrocarbon operations / multiple permits in one area | Gas plant expansion interface areas [CONFIRM] | Conflicting permits or simultaneous incompatible activities (e.g. hot work near venting/testing) create ignition or exposure pathways not visible to any single permit holder. | Fire/explosion, toxic exposure, multiple fatalities; conflicting emergency actions. Persons at risk: All work parties in the shared area; exposure via shared atmosphere, shared escape routes and shared permits. | TO BE VERIFIED: owner SIMOPS procedure and permit register not reviewed — site input required. | 3 x 5 = 15 — HIGH | ENGINEERING: single permit-coordination point with a live permit map/board for the area. ADMIN: SIMOPS matrix of prohibited combinations approved by operations; one accountable Area Authority per area/shift; permit conflict check at issue and at each shift handover; suspension rule when operations activities change. PPE: as per task RA. | Area Authority / Operations Representative Priority: Critical Target date: [SITE INPUT REQUIRED] | 2 x 5 = 10 — HIGH | Owner PTW system; SIMOPS plan; permit register extract Review trigger / reference: Review at each shift change and on any new permit in the area. Ref: owner PTW/SIMOPS procedure [SITE INPUT REQUIRED]. | Evidence status: To be verified — no site evidence reviewed HOLD POINT: hot work prohibited while hydrocarbon venting, draining, sampling or line opening is in progress in the same area unless a specific SIMOPS risk assessment signed by operations authorises it. |

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| RA ID | Work activity / step | Asset / area / workfront | Hazard / hazardous event | Potential consequence & persons at risk | Existing controls | Initial risk (L x S = score) | Additional controls / actions (hierarchy of controls) | Responsible role & target date | Residual risk (L x S = score) | PTW / procedure reference | Verification evidence / status |
|---|---|---|---|---|---|---|---|---|---|---|---|
| RA-03 | Work on or near pressurised hydrocarbon gas / condensate systems (tie-ins, testing, reinstatement) | Compression, processing trains, pipelines within package [CONFIRM] | Uncontrolled release of flammable gas/condensate forming a flammable cloud; jet fire, flash fire or vapour cloud explosion with escalation to adjacent plant. | Multiple fatalities, major asset loss, environmental release; credible major-accident event. Persons at risk: Work crews at the release point, downwind personnel, plant operators; exposure via thermal radiation, blast overpressure and projectiles. | TO BE VERIFIED: fixed gas detection, ESD and fire protection status at the interface not evidenced — site input required. NOTE: a 5x5 matrix alone is not sufficient for this major-accident scenario; HAZID/HAZOP/bow-tie review required (see methodology). | 3 x 5 = 15 — HIGH | ELIMINATION: defer hydrocarbon introduction until construction complete in that system. ENGINEERING: positive isolation, double block and bleed where the owner procedure requires; fixed/portable gas detection coverage verified before work; bonding/earthing. ADMIN: line-break and reinstatement procedure with independent verification; leak test before introduction of hydrocarbons; emergency response on standby for first fill. PPE: FR clothing and personal gas monitor where the site standard requires. | Process Engineer / Area Authority Priority: Critical Target date: [SITE INPUT REQUIRED] | 2 x 5 = 10 — HIGH | Line-opening permit; isolation certificate; leak-test/PSSR documentation Review trigger / reference: Review on any change of system boundary or composition data. Ref: OSHA PSM hazard recognition (technical guidance) [S-14]. | Evidence status: To be verified — no site evidence reviewed HOLD POINT: no introduction of hydrocarbons into new/modified systems without completed leak test, PSSR sign-off and operations acceptance. Residual risk remains HIGH — acceptance only by the authorised owner representative under the client risk matrix. |
| RA-04 | Work where sour gas / H2S may be present in process streams — VERIFICATION-CRITICAL | Any process area — gas composition NOT confirmed in public sources [CONFIRM from process data] | Hydrogen sulfide or other toxic gas exposure if the actual stream composition is sour; public sources do NOT confirm H2S content at Jafurah workfronts. | Acute toxic exposure, single or multiple fatalities, failed rescue attempts by unprotected responders. Persons at risk: Crews opening or working near process systems; exposure via inhalation at the work point and in low-lying areas. | NOT CONFIRMED: no gas composition, P&ID or SDS data available to this pre-assessment. Treat as unassessed until process safety information is supplied. | 3 x 5 = 15 — HIGH | VERIFY FIRST: obtain gas composition/H2S data, hazardous-area drawings and owner gas-testing criteria before any intrusive work. ENGINEERING (if sour service confirmed): fixed H2S detection coverage, ventilation. ADMIN: H2S training and certification for exposed crews, personal H2S monitors with alarm limits set ONLY to the approved site standard (no values assumed here), buddy system, rescue plan with SCBA-equipped responders. PPE: escape respirators/SCBA as the site standard requires. | Process Safety / HSE Manager Priority: Critical Target date: [SITE INPUT REQUIRED] | 2 x 5 = 10 — HIGH | Owner gas-testing procedure; H2S contingency plan [SITE INPUT REQUIRED] Review trigger / reference: Review immediately when composition data is issued. Ref: OSHA PSM hazard recognition (technical guidance) [S-14]. | Evidence status: NOT CONFIRMED — verification-critical HOLD POINT: intrusive work on any process system is prohibited until composition data is received and the H2S controls decision (required / not required) is recorded by a competent person. |

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| RA-05 | Entry/work in enclosed or low-lying areas near process equipment where oxygen deficiency may occur | Pits, sumps, enclosed modules, low points near process areas [CONFIRM] | Oxygen-deficient or gas-enriched atmosphere from leaking inert/hydrocarbon gas accumulating in low or enclosed spaces. | Asphyxiation, fatality, multiple fatalities during attempted rescue. Persons at risk: Workers entering pits/enclosures; maintenance and inspection crews; exposure via displacement of breathable air. | TO BE VERIFIED: site atmosphere-testing regime and area classification not evidenced. | 2 x 5 = 10 — HIGH | ENGINEERING: forced ventilation of enclosed work areas; continuous atmospheric monitoring during occupancy. ADMIN: treat suspect enclosures under the confined-space procedure until classified otherwise; pre-entry testing by an authorised gas tester; attendant posted. PPE: personal multi-gas monitor per site standard. | Authorised Gas Tester / HSE Officer Priority: High Target date: [SITE INPUT REQUIRED] | 1 x 5 = 5 — MODERATE | Confined-space entry permit where classified; gas test certificate Review trigger / reference: Review on any new enclosure or process change. Ref: OSHA Confined Spaces overview (technical guidance) [S-15]. | Evidence status: To be verified — no site evidence reviewed HOLD POINT: no entry where atmospheric test results are outside the limits in the approved site procedure — limits to be taken from that procedure, not assumed. |
| RA-06 | Tie-in and modification work during continued plant operation | Plant interface / tie-in points [CONFIRM] | Unplanned process interruption, control-system upset, spurious trip or emergency shutdown caused by construction activity (signal interference, accidental isolation, ESD activation). | Loss of process control, flaring/upset conditions, knock-on release risk, production and safety-system impairment. Persons at risk: Plant operators and construction crews; exposure via sudden process state change in the work area. | TO BE VERIFIED: operations notification and impairment-management process not reviewed. | 3 x 4 = 12 — HIGH | ADMIN: written operations notification and agreement of work windows; impairment permit and compensating measures for any safety-critical system affected; instrument work only by authorised E&I personnel under operations control; immediate notification protocol to control room before starting and after finishing each task. ENGINEERING: physical protection of live instrument cabling/sensing lines in the work area. | Instrumentation Engineer / Area Authority Priority: High Target date: [SITE INPUT REQUIRED] | 2 x 4 = 8 — MODERATE | Impairment permit; E&I work authorisation; control-room notification log Review trigger / reference: Review on any ESD/alarm event in the area. Ref: owner MOC/impairment procedure [SITE INPUT REQUIRED]. | Evidence status: To be verified — no site evidence reviewed HOLD POINT: no work on or near live instrument/ESD systems without an approved impairment plan and control-room acknowledgement. |
| RA-07 | Temporary bypass, override or alarm impairment for construction/commissioning convenience | Any safety-critical system in the interface area [CONFIRM] | Safety-critical protection (gas detection, ESD, relief, fire systems) left bypassed or impaired without management of change, removing a barrier during high-risk work. | Undetected release escalation, major fire/explosion, multiple fatalities. Persons at risk: All personnel in the affected area; exposure via loss of the detection/protection barrier. | TO BE VERIFIED: no bypass register reviewed; site input required. | 2 x 5 = 10 — HIGH | ADMIN: bypasses only under formal MOC/impairment permit with named owner, time limit, compensating measures (e.g. continuous manned monitoring, portable detection) and restoration verification; bypass register reviewed at each shift handover; overdue bypass escalation to plant management. ENGINEERING: restore and function-test the system before the associated hot/intrusive work resumes. | Process Safety Engineer Priority: Critical Target date: [SITE INPUT REQUIRED] | 1 x 5 = 5 — MODERATE | MOC / impairment permit; bypass register Review trigger / reference: Review at every shift handover while any bypass is active. Ref: OSHA PSM overview (technical guidance) [S-13]. | Evidence status: To be verified — no site evidence reviewed HOLD POINT: hot work and line opening prohibited in any area where its credited detection/protection barrier is bypassed without approved compensating measures in place. |

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| RA-08 | Commissioning and first introduction of hydrocarbons / energisation of new systems | New processing trains, compression and utilities in package [CONFIRM] | Leaks at flanges/joints not yet proven in service, gasket failure, or incorrect lineup during first fill and pressurisation. | Flammable/toxic release, fire, fatality, damage to new and operating plant. Persons at risk: Commissioning team, operations staff, nearby construction crews; exposure at leak points and along pressurised systems. | TO BE VERIFIED: commissioning procedures and PSSR process are owner documents — not reviewed. | 3 x 5 = 15 — HIGH | ENGINEERING: staged pressurisation with hold points and leak surveys at each stage; remote monitoring where practicable. ADMIN: PSSR completed and signed before introduction; exclusion zone around the system during first fill sized by the commissioning procedure; joint construction/commissioning boundary control with a single accountable authority; all non-essential personnel cleared from the area. PPE: per commissioning procedure. | Commissioning Lead Priority: Critical Target date: [SITE INPUT REQUIRED] | 2 x 5 = 10 — HIGH | PSSR certificate; commissioning permit; operations acceptance certificate Review trigger / reference: Review at each commissioning stage gate. Ref: OSHA PSM overview (technical guidance) [S-13]. | Evidence status: To be verified — no site evidence reviewed HOLD POINT: no hydrocarbon introduction without PSSR closure, leak-test records and signed operations acceptance. |
| RA-09 | Isolation of mechanical / electrical / instrument energy for intrusive maintenance and tie-in work | All systems in package [CONFIRM isolation points from P&IDs] | Failure to identify and isolate all energy sources (pressure, electrical, stored, gravity, chemical) — unexpected start-up or release during work. | Fatality or serious injury from release of pressure, moving equipment, electric shock or falling load. Persons at risk: Fitters, electricians, instrument technicians at the work point; exposure via direct energy release. | TO BE VERIFIED: owner LOTO/isolation procedure and isolation hardware availability not evidenced. | 3 x 5 = 15 — HIGH | ENGINEERING: positive isolation (physical disconnection/blinding) for line breaking per owner procedure; lockable isolation points. ADMIN: isolation certificate listing every energy source, applied by an authorised isolator; personal locks and tags for each worker; zero-energy verification (try-start, test-for-dead, pressure bleed check) recorded before work; group lockout for multi-craft work. PPE: task PPE. | Authorised Isolator / Area Authority Priority: Critical Target date: [SITE INPUT REQUIRED] | 1 x 5 = 5 — MODERATE | Isolation certificate; LOTO register; PTW Review trigger / reference: Review on any isolation boundary change. Ref: owner isolation/LOTO procedure [SITE INPUT REQUIRED]. | Evidence status: To be verified — no site evidence reviewed HOLD POINT: work must not start until zero-energy state is proven and recorded; any doubt about an energy source stops the job. |
| RA-10 | Line opening / flange breaking on process piping | Tie-in and modification points [CONFIRM from line list] | Residual pressure, trapped gas/liquid or incorrect line identification releases hydrocarbons or chemicals at the break point; line-of-fire ejection of components. | Fire, toxic exposure, fatality, environmental release. Persons at risk: Pipefitters at the flange face and personnel in the line of fire; exposure via jet release and spray. | TO BE VERIFIED: owner line-opening procedure, drain/vent provisions and lineup drawings not reviewed. | 4 x 5 = 20 — EXTREME | ELIMINATION: break lines only at designated break points in the approved procedure. ENGINEERING: drain, vent, purge and prove the line depressurised at the break point before loosening; first-break technique from a protected position (loosen far side first, stand clear of the line of fire). ADMIN: line identification independently verified against the line list/P&ID by two competent persons; gas test at the break point; line-opening permit signed by operations. PPE: face/chemical protection per the fluid SDS and site standard. | Mechanical Supervisor / Area Authority Priority: Critical Target date: [SITE INPUT REQUIRED] | 2 x 5 = 10 — HIGH | Line-opening permit; isolation certificate; gas test certificate Review trigger / reference: Review for every new line break. Ref: owner line-opening procedure [SITE INPUT REQUIRED]; OSHA PSM hazard recognition [S-14]. | Evidence status: To be verified — no site evidence reviewed HOLD POINT: first break prohibited until drain/vent proven, gas test satisfactory and permit signed. Stop immediately on any unexpected pressure/product at the break. |

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| RA-11 | Draining, venting, blinding/spading and proving isolation before intrusive work | Process systems in package [CONFIRM] | Incorrect blind location, passing isolation valve, or unrecorded blind left in / removed early defeats the isolation. | Release of hydrocarbons into the work area during intrusive work; fatality; or process incident on reinstatement. Persons at risk: Intrusive-work crews relying on the isolation; operators on reinstatement. | TO BE VERIFIED: blind register and marking practice not evidenced. | 3 x 5 = 15 — HIGH | ADMIN: blind/spade register with unique tags, location marked on P&ID, installed/removed only by authorised persons with independent verification and sign-off; isolation proving (bleed-point check) recorded on the isolation certificate; reinstatement checklist cross-checked against the register before repressurisation. ENGINEERING: blinds rated for the system design conditions per the owner specification (rating to be confirmed from documents — not assumed here). | Authorised Isolator Priority: Critical Target date: [SITE INPUT REQUIRED] | 1 x 5 = 5 — MODERATE | Isolation certificate; blind register Review trigger / reference: Review on every register change. Ref: owner isolation procedure [SITE INPUT REQUIRED]. | Evidence status: To be verified — no site evidence reviewed HOLD POINT: no repressurisation until the blind register is reconciled and signed by operations. |
| RA-12 | Pipeline tie-in and reinstatement (welding, bolting, gasket replacement) | Pipeline / piping tie-in points [CONFIRM] | Joint integrity failure from incorrect gasket, bolt torque, alignment or weld quality, discovered only under operating pressure. | Delayed hydrocarbon release in service, fire, fatality, unplanned shutdown. Persons at risk: Operations personnel and future maintenance crews; public/process exposure via in-service leak. | TO BE VERIFIED: WPS/PQR, bolting procedure and NDT scope are package documents — not reviewed. | 3 x 4 = 12 — HIGH | ENGINEERING: qualified WPS and welders for the joint class; controlled bolting with calibrated torque/tension equipment and recorded values; specified gasket type verified at fit-up. ADMIN: joint register with hold-point inspection by QA/QC before insulation/closure; NDT extent per the approved ITP; reinstatement leak test before handover. PPE: task PPE. | QA/QC Inspector / Mechanical Engineer Priority: High Target date: [SITE INPUT REQUIRED] | 1 x 4 = 4 — LOW | Approved WPS; ITP; joint register; leak-test record Review trigger / reference: Review on any joint repair/rework. Ref: approved project ITP [SITE INPUT REQUIRED]. | Evidence status: To be verified — no site evidence reviewed HOLD POINT: no joint to be insulated, buried or closed out before QA/QC hold-point release is signed. |
| RA-13 | Confined-space entry into vessels, columns, tanks, sumps or large-bore equipment | Equipment in package requiring internal work [CONFIRM entry list] | Toxic/flammable/ oxygen-deficient atmosphere, engulfment, or heat stress inside the space; failed rescue multiplying casualties. | Single or multiple fatalities (entrant and would-be rescuers). Persons at risk: Entrants, attendant, rescue team; exposure via atmosphere inside the space. | TO BE VERIFIED: owner confined-space procedure, equipment classification and rescue capability not evidenced. | 3 x 5 = 15 — HIGH | ELIMINATION: use remote inspection/cleaning methods where practicable to avoid entry. ENGINEERING: isolation and blinding of all connected lines, cleaning/purging proven by testing, continuous ventilation, continuous atmospheric monitoring. ADMIN: entry permit with named attendant, communications check, entry log, and a rehearsed rescue plan with equipment at the entry point before entry starts; entrant/attendant training verified. PPE: respiratory protection per test results and site procedure. | Confined-Space Entry Supervisor Priority: Critical Target date: [SITE INPUT REQUIRED] | 1 x 5 = 5 — MODERATE | Confined-space entry permit; gas test certificate; rescue plan Review trigger / reference: Review for every entry. Ref: OSHA Confined Spaces overview (technical guidance) [S-15]; owner confined-space procedure [SITE INPUT REQUIRED]. | Evidence status: To be verified — no site evidence reviewed HOLD POINT: no entry without a rescue capability physically in place; entry stops immediately on monitor alarm, ventilation failure or attendant leaving post. |

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| RA-14 | Nitrogen / inert-gas purging of systems for commissioning or maintenance — VERIFICATION-CRITICAL (only if N2 is used) | Systems under purge in package [CONFIRM whether N2 is used] | Nitrogen leakage or discharge into work areas creating an oxygen-deficient atmosphere without warning properties (colourless, odourless). | Rapid asphyxiation, fatalities including rescuers. Persons at risk: Personnel near purge vents, open ends and low areas; exposure via inhalation of oxygen-depleted air. | NOT CONFIRMED: use of nitrogen at the workfront is not evidenced in public sources — confirm from the commissioning procedure. | 2 x 5 = 10 — HIGH | VERIFY FIRST: confirm purging medium and procedure. ENGINEERING: purge discharge routed to a safe location; barricaded vent points. ADMIN: purge plan with marked exclusion zones, oxygen monitoring at access points, warning signage in workforce languages, prohibition on entering purged systems/spaces until oxygen proven normal by test. PPE: personal O2 monitor where the procedure requires. | Commissioning Lead Priority: Critical Target date: [SITE INPUT REQUIRED] | 1 x 5 = 5 — MODERATE | Commissioning/ purge procedure; gas test records Review trigger / reference: Review at each purge operation. Ref: commissioning procedure [SITE INPUT REQUIRED]. | Evidence status: NOT CONFIRMED — verification-critical HOLD POINT: no entry into, or breaking into, any purged system until oxygen content is proven acceptable by test and recorded. |
| RA-15 | Electrical work on new and existing installations; temporary supplies and backfeed risks | Substations, MCCs, field installations in package [CONFIRM voltages from drawings] | Contact with live parts, backfeed from temporary generators/UPS/interconnections, or incorrect isolation of multi-source equipment. | Electrocution, arc-flash burns, fatality. Persons at risk: Electricians and instrument technicians; exposure via direct contact and arc flash. | TO BE VERIFIED: electrical safety rules, authorised-person register and drawings not reviewed. Voltages are NOT stated in this assessment — obtain from approved drawings. | 3 x 5 = 15 — HIGH | ENGINEERING: isolation at all sources including temporary supplies, with interlocking/earthing where the rules require. ADMIN: work only by electrically authorised persons; prove dead with a verified tester before touching; electrical permit for HV work as defined by the site rules; temporary-supply register and RCD protection for portable equipment. PPE: arc-rated clothing and voltage-rated gloves per the site electrical standard. | Authorised Electrical Person / E&I Engineer Priority: Critical Target date: [SITE INPUT REQUIRED] | 1 x 5 = 5 — MODERATE | Electrical permit / sanction for test; isolation certificate Review trigger / reference: Review on any supply configuration change. Ref: owner electrical safety rules [SITE INPUT REQUIRED]. | Evidence status: To be verified — no site evidence reviewed HOLD POINT: no contact work until prove-dead is recorded; any unidentified cable is treated as live and stops the job. |
| RA-16 | Pre-startup safety review (PSSR) and handover of completed systems to operations | Completed trains/systems in package [CONFIRM] | Construction debris, open punch items, missing blinds/valve line-up errors or incomplete documentation transferred to operations at handover. | Process incident at start-up, release, injury, asset damage. Persons at risk: Operations and commissioning teams at start-up; exposure via unexpected system behaviour. | TO BE VERIFIED: handover/PSSR procedure and punch-list system not reviewed. | 3 x 4 = 12 — HIGH | ADMIN: system-by-system PSSR with operations, construction, process and HSE sign-off; punch items categorised — no start-up with open safety-critical items; valve/line-up verification walkdown against P&IDs; as-built and test documentation complete before acceptance; formal boundary certificate defining which party controls each system. PPE: n/a (administrative control point). | Commissioning Lead / Operations Representative Priority: High Target date: [SITE INPUT REQUIRED] | 1 x 4 = 4 — LOW | PSSR certificate; handover certificate; punch list Review trigger / reference: Review at every system handover. Ref: owner PSSR procedure [SITE INPUT REQUIRED]. | Evidence status: To be verified — no site evidence reviewed HOLD POINT: operations acceptance is the gate — construction permits close and operations permits govern from the recorded transfer time. |

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| RA-17 | Heavy and critical crane lifts of equipment, modules and spools | Laydown areas and plant installation points [CONFIRM] | Rigging failure, incorrect load weight/centre of gravity, or crane malfunction causing a dropped or swinging load. | Fatality, serious injury, damage to live plant with secondary hydrocarbon release. Persons at risk: Rigging crew and personnel in the drop zone; exposure via being under or near the suspended load path. | TO BE VERIFIED: lifting procedure, crane/rigging certification records and appointed-person arrangements not reviewed. | 3 x 5 = 15 — HIGH | ENGINEERING: lift only with certified crane and rigging matched to a calculated lift plan (load weight and CoG from vendor/drawing data — never estimated); tag lines for load control. ADMIN: lifts categorised (routine/complex/critical) per the owner lifting standard with an approved lift plan and appointed person for complex/critical lifts; barricaded drop zone; trial lift and brake hold check; no personnel under suspended loads — enforced by the lift supervisor. PPE: task PPE. | Lifting Supervisor / Appointed Person Priority: Critical Target date: [SITE INPUT REQUIRED] | 1 x 5 = 5 — MODERATE | Approved lift plan; crane/rigging certificates; lifting permit where required Review trigger / reference: Review for every complex/critical lift. Ref: owner lifting procedure [SITE INPUT REQUIRED]. | Evidence status: To be verified — no site evidence reviewed HOLD POINT: critical lifts over or near live plant require operations approval in addition to the lift plan; lift stops on any rigging defect, alarm or loss of communication. |
| RA-18 | Crane set-up and operation on desert ground conditions | Crane pads / work areas [CONFIRM ground data] | Ground bearing failure, outrigger sinking into sand/uncompacted fill, or buried-service collapse under outrigger loads causing crane overturn. | Fatality, major equipment loss, impact with live plant. Persons at risk: Crane crew and nearby workers; exposure via overturn radius and load drop. | TO BE VERIFIED: geotechnical/ground-bearing data, compaction records and underground-service drawings not reviewed. | 3 x 5 = 15 — HIGH | ENGINEERING: engineered crane pads/mats sized from a ground-bearing assessment by a competent person; outrigger loads calculated in the lift plan. ADMIN: underground-service clearance for the set-up area before positioning; daily ground-condition check and after-rain re-verification; set-up sign-off by the appointed person before first lift. PPE: task PPE. | Appointed Person (Lifting) Priority: Critical Target date: [SITE INPUT REQUIRED] | 1 x 5 = 5 — MODERATE | Lift plan with ground assessment; excavation/utility clearance for pad area Review trigger / reference: Review after rain, groundworks or relocation. Ref: owner lifting procedure [SITE INPUT REQUIRED]. | Evidence status: To be verified — no site evidence reviewed HOLD POINT: no lift from any set-up position without signed ground/set-up verification; any visible settlement stops lifting immediately. |
| RA-19 | Lifting and work at height during high wind, sandstorm or lightning conditions | All outdoor workfronts | Loss of load control, scaffold/access instability, or lightning strike when work continues beyond approved weather limits. | Dropped load, falls, electrocution; multiple injuries possible. Persons at risk: Lifting crews, scaffolders, work-at-height personnel; exposure via weather at exposed locations. | TO BE VERIFIED: site-specific weather limits and monitoring arrangements not evidenced. NOTE: KAEFER's published project story records heavy rain, strong winds and sandstorms as real challenges at Jafurah [S-05]. | 3 x 4 = 12 — HIGH | ADMIN: apply the wind/weather limits in the crane manufacturer's manual and the approved site procedure (numeric limits to be taken from those documents — none assumed here); anemometer readings at the workfront; defined stop-work triggers for lightning proximity and sandstorm visibility, with a named role authorised and required to call the stop; secure loose materials ahead of forecast storms. ENGINEERING: load control with tag lines; scaffold ties per design. | Construction Manager / Lifting Supervisor Priority: High Target date: [SITE INPUT REQUIRED] | 2 x 4 = 8 — MODERATE | Site weather-monitoring log; lift plan weather criteria Review trigger / reference: Review each shift in the storm season. Ref: crane manual + site procedure [SITE INPUT REQUIRED]; KAEFER project story [S-05]. | Evidence status: To be verified — no site evidence reviewed STOP-WORK: any worker may stop work on lightning or sudden sandstorm; resumption only after the named supervisor confirms conditions are within limits. |

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| RA-20 | Scaffold erection, alteration and dismantling (including suspended/underhung scaffolds) | Access locations in package [CONFIRM] | Scaffold collapse from incorrect erection/ties/loading, or falls during erection and dismantling. | Fatality, serious injury, falling materials onto workers below. Persons at risk: Scaffolders, scaffold users, workers below; exposure via falls and falling components. | TO BE VERIFIED: scaffold design standard, inspection/tag system and scaffolder competency records not reviewed. KAEFER's published scope at Jafurah Phase 1 included access/scaffolding incl. suspended types [S-05] — current package scope to be confirmed. | 3 x 5 = 15 — HIGH | ENGINEERING: scaffolds erected to an approved design/standard with ties, bracing and loading limits; design check for suspended/special scaffolds. ADMIN: erection/alteration only by competent scaffolders under a scaffold supervisor; inspection and tag (safe/unsafe) before first use, weekly and after alteration or severe weather; incomplete scaffolds physically barred and signed. PPE: fall-arrest with verified anchor during erection where the method requires. | Scaffold Supervisor Priority: Critical Target date: [SITE INPUT REQUIRED] | 1 x 5 = 5 — MODERATE | Scaffold inspection register and tag; design drawings for special scaffolds Review trigger / reference: Review weekly and after any storm. Ref: owner scaffold standard [SITE INPUT REQUIRED]. | Evidence status: To be verified — no site evidence reviewed HOLD POINT: no access to any scaffold without a current 'safe for use' tag; red-tagged scaffolds are physically closed. |
| RA-21 | Work at height on structures, platforms, roofs and open edges; dropped-object exposure | Elevated work areas in package [CONFIRM] | Fall from height through open edges, penetrations or brittle surfaces; tools/materials dropped onto personnel below. | Fatality or life-changing injury to the faller or the person struck. Persons at risk: Fitters, welders, inspectors at height; all personnel at lower levels. | TO BE VERIFIED: edge-protection status, anchor certification and rescue arrangements not evidenced. | 4 x 5 = 20 — EXTREME | ELIMINATION: pre-assemble at grade where practicable. ENGINEERING: collective protection first — guardrails, covered/marked penetrations, certified anchor points; tool tethering and toe boards; barricaded drop zones below. ADMIN: work-at-height permit where the site system requires; 100% tie-off rule per site standard with inspected harnesses; rescue-from-height plan and equipment at the workfront before work starts. PPE: full-body harness with double lanyard per site standard. | Construction Supervisor / HSE Officer Priority: Critical Target date: [SITE INPUT REQUIRED] | 2 x 5 = 10 — HIGH | Work-at-height permit (if required by site system); rescue plan Review trigger / reference: Review for every new elevated workfront. Ref: owner work-at-height procedure [SITE INPUT REQUIRED]. | Evidence status: To be verified — no site evidence reviewed HOLD POINT: work at unprotected edges prohibited; any fall-arrest deployment or anchor defect stops the task pending review. |
| RA-22 | Transport and lift-in of prefabricated scaffold modules / large preassemblies — CONDITIONAL (only if this method is in the current package) | Fabrication yard to plant route [CONFIRM — method reported historically, not confirmed current] | Load shift or instability during trailer transport and crane lift-in of large prefabricated modules on internal roads shared with plant traffic. | Load drop, vehicle collision, fatality, damage to plant. Persons at risk: Transport and lifting crews, other road users; exposure along the transport route and at lift-in. | HISTORICALLY REPORTED ONLY: KAEFER described modular scaffolding assembled in a yard about 1 km away and lifted/trailered to site at Jafurah Phase 1 [S-05]. NOT confirmed as a current method — do not plan on it without the current method statement. | 3 x 5 = 15 — HIGH | VERIFY FIRST: confirm the method statement for the current package. ENGINEERING: engineered transport frame/lashing and lift points for each module type. ADMIN: route survey and agreed movement windows coordinated with plant operations; escort vehicles; lift-in under RA-17 controls; exclusion of other traffic during movement. PPE: task PPE. | Construction Manager Priority: High Target date: [SITE INPUT REQUIRED] | 1 x 5 = 5 — MODERATE | Method statement; transport plan; lift plan Review trigger / reference: Review on any route or module change. Ref: KAEFER project story [S-05] (historical method only). | Evidence status: Historically reported only — confirm current HOLD POINT: no module movement without an approved transport and lift-in plan accepted by operations for the route through live areas. |

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| RA-23 | Use of mobile elevating work platforms (MEWPs) and suspended access equipment | Elevated access locations [CONFIRM] | MEWP overturn on soft/uneven ground, ejection or entrapment of occupants, contact with overhead structures or live equipment. | Fatality or serious injury to occupants and ground crew. Persons at risk: MEWP operators/occupants and ground personnel; exposure at height and in the overturn radius. | TO BE VERIFIED: operator certification, machine inspection status and ground rules not evidenced. | 3 x 5 = 15 — HIGH | ENGINEERING: machines selected for terrain and reach, inspected and certified; ground assessed before positioning (RA-18 principles). ADMIN: trained/authorised operators only; harness use in boom-type platforms per site standard; pre-use checks recorded; exclusion zone and ground spotter for movements near plant; emergency lowering procedure briefed and rescue plan in place. PPE: harness per site standard. | Construction Supervisor Priority: High Target date: [SITE INPUT REQUIRED] | 1 x 5 = 5 — MODERATE | Equipment certificates; operator authorisation records Review trigger / reference: Review on every machine change. Ref: manufacturer manual; owner procedure [SITE INPUT REQUIRED]. | Evidence status: To be verified — no site evidence reviewed HOLD POINT: defective or uncertified machines are tagged out; no operation outside the manufacturer's wind limit (from the machine manual — not assumed here). |
| RA-24 | Excavation and trenching for foundations, pipelines and underground services | Civil work areas in package [CONFIRM] | Trench/excavation wall collapse; strike on buried live services (cables, pipelines, fibre) not shown or incorrectly marked. | Burial/asphyxiation fatality; electrocution or hydrocarbon release from a service strike. Persons at risk: Workers in and beside the excavation; plant operators; exposure via collapse zone and service strike point. | TO BE VERIFIED: underground-service drawings, permit practice and soil data not reviewed. | 3 x 5 = 15 — HIGH | ENGINEERING: protective system (benching/shoring/shielding) selected by a competent person for the actual soil and depth — dimensions from the excavation design, not assumed here; service detection scan and trial holes to prove buried-service positions before mechanical digging. ADMIN: excavation permit with services clearance signed by operations/E&I; daily and after-rain inspections by the competent person; mechanical excavation stops at the agreed distance from marked services and hand digging takes over. PPE: task PPE. | Civil Supervisor / Competent Person (Excavations) Priority: Critical Target date: [SITE INPUT REQUIRED] | 1 x 5 = 5 — MODERATE | Excavation permit; services-clearance drawings; scan records Review trigger / reference: Review daily and after rain or design change. Ref: owner excavation procedure [SITE INPUT REQUIRED]. | Evidence status: To be verified — no site evidence reviewed HOLD POINT: no mechanical excavation without signed services clearance; any unmarked service found stops work in that excavation. |
| RA-25 | Work in and around open excavations — access, egress, edge protection and surcharge | Excavation locations [CONFIRM] | Falls into excavations by workers or public/plant personnel; spoil, materials or mobile plant surcharging the edge causing collapse. | Fractures, fatality, burial. Persons at risk: All personnel and vehicle operators near excavations; exposure via unprotected edges and access points. | TO BE VERIFIED: edge-protection and access provisions not evidenced. | 3 x 4 = 12 — HIGH | ENGINEERING: rigid edge barriers around open excavations, safe crossing points, ladders/access at the intervals required by the site standard. ADMIN: spoil and plant kept back at the distance set by the excavation design; excavations left open overnight are barricaded, signed and lit; inspections recorded daily. PPE: task PPE. | Civil Supervisor Priority: High Target date: [SITE INPUT REQUIRED] | 1 x 4 = 4 — LOW | Excavation permit; inspection records Review trigger / reference: Review daily while excavations remain open. Ref: owner excavation procedure [SITE INPUT REQUIRED]. | Evidence status: To be verified — no site evidence reviewed STOP-WORK: any excavation found with failed edge protection or undermined access is closed until corrected. |

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| RA-26 | Hydrostatic pressure testing of piping and equipment | Test boundaries in package [CONFIRM test pressures from approved test procedure] | Failure of a test blank, hose, fitting or weak component under test pressure releasing a high-energy water jet or projectile. | Fatality or serious injury from projectiles/jet; equipment damage. Persons at risk: Test crew and anyone inside the test boundary; exposure via line of fire of pressurised components. | TO BE VERIFIED: test procedures, calibrated gauge records and boundary practice not reviewed. Test pressures are NOT stated in this assessment — they must come from the approved test pack. | 3 x 5 = 15 — HIGH | ENGINEERING: test pack engineered and approved (test medium, pressure, hold points, vent/drain points) by a competent engineer; restraints on temporary hoses/connections; calibrated gauges with current certificates. ADMIN: barricaded exclusion zone around the full test boundary during pressurisation and hold; pressurisation in stages with inspection at each stage; no one inside the boundary while pressure is above the level set in the test procedure; controlled depressurisation and drain-down. PPE: task PPE outside the line of fire. | Mechanical / Commissioning Engineer Priority: Critical Target date: [SITE INPUT REQUIRED] | 1 x 5 = 5 — MODERATE | Approved test pack; pressure-test permit; calibration certificates Review trigger / reference: Review for every test. Ref: approved test procedure [SITE INPUT REQUIRED]. | Evidence status: To be verified — no site evidence reviewed HOLD POINT: pressurisation only under the test supervisor's single command with the boundary confirmed clear and recorded. |
| RA-27 | Pneumatic pressure testing — CONDITIONAL (only where specified and justified) | Test boundaries in package [CONFIRM] | Stored-energy release on component failure is far more violent with gas than with liquid; blast and projectile effects over a wide area. | Multiple fatalities possible; major equipment damage. Persons at risk: Test crew and personnel well beyond the immediate boundary; exposure via blast/projectile range. | NOT CONFIRMED: pneumatic testing is not evidenced in public sources for this package — include only where the approved design/test specification requires it. | 2 x 5 = 10 — HIGH | ELIMINATION/ SUBSTITUTION: use hydrostatic testing wherever the specification allows. ENGINEERING/ADMIN (where pneumatic testing is unavoidable): specific engineered risk review and written justification approved at project-management level; enlarged exclusion zone sized by the review; staged pressurisation with remote monitoring and no personnel in the zone during pressure steps; pre-test NDT/inspection of joints per the test pack. PPE: n/a inside exclusion zone (exclusion is the control). | Mechanical / Commissioning Engineer Priority: Critical Target date: [SITE INPUT REQUIRED] | 1 x 5 = 5 — MODERATE | Engineered pneumatic test procedure with management approval; pressure-test permit Review trigger / reference: Review for every pneumatic test. Ref: approved test specification [SITE INPUT REQUIRED]. | Evidence status: NOT CONFIRMED — verification-critical HOLD POINT: pneumatic testing must not proceed without the specific engineered review and senior approval recorded in the test pack. |

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| RA-28 | Pipeline welding, fit-up, coating/wrapping and inspection | Pipeline spreads and tie-in locations [CONFIRM] | Welding arc/fire in an open spread, coating material hazards, manual handling of pipe and fit-up pinch points; joint failure if quality controls fail. | Burns, fire, crush injuries; in-service joint failure (see RA-12). Persons at risk: Welders, fitters, coating crews, inspectors; exposure at the joint and during pipe handling. | TO BE VERIFIED: WPS, coating SDS and spread method statements not reviewed. | 3 x 4 = 12 — HIGH | ENGINEERING: pipe handling with certified slings/spreaders and side-boom/pipe-layer controls per the method statement; welding screens and fire prevention at each station. ADMIN: welder qualification verified against the WPS; coating products used only after SDS review with controls implemented (ventilation, PPE, storage); fit-up clamps and line-up controls to remove pinch-point exposure; joint quality per RA-12. PPE: welding and coating PPE per SDS/site standard. | Welding Supervisor / QA-QC Inspector Priority: High Target date: [SITE INPUT REQUIRED] | 2 x 4 = 8 — MODERATE | Approved WPS; SDS file; method statement Review trigger / reference: Review on any material or WPS change. Ref: approved WPS/ITP [SITE INPUT REQUIRED]. | Evidence status: To be verified — no site evidence reviewed HOLD POINT: no coating product to be opened on site before its SDS is on file and briefed to the crew. |
| RA-29 | Desert ground conditions, dust and remote-location logistics for civil and pipeline work | Remote spreads, access roads and work areas [CONFIRM] | Soft sand, washouts after rain, dust-reduced visibility and long distances to medical care degrading both routine work and emergency response. | Vehicle incidents, delayed treatment of injuries/heat illness increasing severity, work errors in poor visibility. Persons at risk: All field crews, especially drivers and lone/remote work parties; exposure via travel and distance from help. | HISTORICALLY REPORTED: remoteness, heat, wind and heavy rain at Jafurah are recorded in KAEFER's project story [S-05]; current logistics plan not reviewed. | 3 x 4 = 12 — HIGH | ENGINEERING: maintained and graded access routes; water/fuel reserves at remote fronts. ADMIN: journey management for remote movements (RA-36); check-in/check-out system for remote parties with escalation if a check-in is missed; work stops when visibility falls below the level set in the site procedure; emergency rendezvous points agreed with plant emergency services. PPE: task PPE plus dust protection in storms. | Construction Manager / Logistics Coordinator Priority: High Target date: [SITE INPUT REQUIRED] | 2 x 4 = 8 — MODERATE | Journey management plan; site logistics plan Review trigger / reference: Review weekly and on route changes. Ref: site logistics plan [SITE INPUT REQUIRED]; KAEFER [S-05]. | Evidence status: Historically reported only — confirm current STOP-WORK: remote parties stop and shelter on loss of communications or severe weather warning until contact is restored. |

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| RA-30 | Hot work: welding, cutting, grinding in or near process areas | Fabrication areas and in-situ hot work locations [CONFIRM] | Sparks/slag igniting flammable atmosphere, combustible materials or hydrocarbon residues; fire spreading beyond the work point. | Fire/explosion, fatalities, major plant damage. Persons at risk: Hot-work crew, fire watch, nearby workers and plant; exposure via ignition at and beyond the immediate work point. | TO BE VERIFIED: owner hot-work procedure, gas-testing criteria and fire-watch practice not reviewed. | 3 x 5 = 15 — HIGH | ELIMINATION: prefabricate in a designated safe fabrication area wherever possible instead of in-situ hot work. ENGINEERING: spark containment (blankets/screens), removal/isolation of combustibles within the radius set by the site procedure. ADMIN: hot-work permit with pre-work and continuous gas testing where the procedure requires, dedicated trained fire watch during work and for the post-work fire-check period specified in the site procedure, suitable extinguishers at the point of work verified before starting. PPE: welding PPE per site standard. | Hot Work Supervisor / Fire Watch Priority: Critical Target date: [SITE INPUT REQUIRED] | 1 x 5 = 5 — MODERATE | Hot-work permit; gas test certificate; fire-watch record Review trigger / reference: Review for every hot-work permit. Ref: OSHA welding/cutting guidance (technical guidance) [S-16]; owner hot-work procedure [SITE INPUT REQUIRED]. | Evidence status: To be verified — no site evidence reviewed HOLD POINT: hot work stops immediately on gas-test alarm, loss of fire watch, or any plant upset alarm in the area. |
| RA-31 | Welding and thermal cutting — fume, radiation and burn exposure | All welding locations | Welding fume and metal-fume inhalation, UV/IR eye and skin damage ('arc eye', burns), contact burns from hot metal. | Occupational lung disease over time, eye injury, burns; severity increases in confined/enclosed spaces. Persons at risk: Welders and nearby helpers/bystanders; exposure via fume plume and arc radiation. | TO BE VERIFIED: ventilation provisions, exposure monitoring and welder health surveillance not evidenced. | 4 x 3 = 12 — HIGH | ENGINEERING: local exhaust ventilation or fume extraction for enclosed/semi-enclosed welding; welding screens to protect bystanders. ADMIN: fume exposure assessment for the consumables in use (from SDS) and monitoring where the assessment requires; hot-work pieces marked/quarantined until cool; rotation to limit individual exposure where monitoring indicates. PPE: welding helmet with correct shade, FR clothing, and respiratory protection where ventilation cannot control fume (selected from the exposure assessment). | HSE Officer / Welding Supervisor Priority: Moderate Target date: [SITE INPUT REQUIRED] | 2 x 3 = 6 — MODERATE | SDS file for consumables; exposure assessment Review trigger / reference: Review on consumable or location change. Ref: OSHA welding hazards (technical guidance) [S-16]. | Evidence status: To be verified — no site evidence reviewed STOP-WORK: welding in enclosed spaces stops if ventilation fails or fume monitoring exceeds the action level in the site hygiene plan. |

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| RA-32 | Temporary power distribution for construction (generators, distribution boards, cables) | All construction areas [CONFIRM] | Damaged/trailing cables, overloaded circuits, missing RCD/earth protection in dusty or wet conditions causing shock or fire. | Electrocution, burns, fire damaging plant or materials. Persons at risk: All site personnel, especially trades using portable tools; exposure via contact with damaged equipment/cables. | TO BE VERIFIED: temporary-electrical installation standard and inspection regime not reviewed. | 3 x 4 = 12 — HIGH | ENGINEERING: industrial-grade distribution with RCD/earth-leakage protection, armoured or protected cable runs, cables elevated or ramped at crossings. ADMIN: installation and alteration only by authorised electricians; inspection and tagging regime for portable tools and leads with defective items removed immediately; generators sited, earthed, refuelled under a safe procedure away from hot work. PPE: task PPE. | E&I Engineer / Authorised Electrical Person Priority: High Target date: [SITE INPUT REQUIRED] | 1 x 4 = 4 — LOW | Temporary electrical installation certificates; inspection register Review trigger / reference: Review monthly and after any electrical incident. Ref: owner electrical rules [SITE INPUT REQUIRED]. | Evidence status: To be verified — no site evidence reviewed STOP-WORK: any damaged cable or defeated protective device takes the circuit out of service until repaired by an authorised electrician. |
| RA-33 | Instrument loop testing, energisation and rotating-equipment testing during commissioning | Systems under commissioning [CONFIRM] | Unexpected energisation or rotation of equipment while construction personnel still work on connected systems at the construction/commissioning boundary. | Fatality or serious injury from rotating equipment, pressure or electrical energy. Persons at risk: Construction crews on connected systems and commissioning technicians; exposure via shared systems across the boundary. | TO BE VERIFIED: commissioning boundary control and system-status board not evidenced. | 3 x 5 = 15 — HIGH | ADMIN: formal system-boundary certificates (see RA-16) with a single status board/register; energisation notices issued to all affected parties a defined period before energisation; physical warning tags/signs on energised systems; permit systems of construction and commissioning cross-checked daily at the boundary meeting. ENGINEERING: guards refitted before rotation tests; remote start where the design allows. | Commissioning Lead Priority: Critical Target date: [SITE INPUT REQUIRED] | 1 x 5 = 5 — MODERATE | Energisation notice; boundary certificate; commissioning permit Review trigger / reference: Review daily during commissioning phase. Ref: commissioning procedure [SITE INPUT REQUIRED]. | Evidence status: To be verified — no site evidence reviewed HOLD POINT: no energisation of any system with open construction permits on that system — the boundary meeting must confirm permit closure first. |
| RA-34 | NDT / industrial radiography — CONDITIONAL (only if radiography is in the package) | Weld inspection locations [CONFIRM] | Ionising radiation exposure from an unshielded or uncontrolled source; source loss or failure to retract. | Radiation overexposure of radiography crew or unknowing bystanders; regulatory incident. Persons at risk: Radiography crew and any person entering the controlled area; exposure via proximity to the exposed source. | NOT CONFIRMED: use of radiography at the workfront is not evidenced in public sources — confirm from the inspection plan/ITP. | 2 x 4 = 8 — MODERATE | ENGINEERING: collimation and shielding to minimise the controlled area. ADMIN: radiography only by licensed/authorised personnel under the applicable Saudi regulatory requirements and site procedure (licensing route to be confirmed — not assumed here); calculated and barricaded controlled area with warning signs/lights in workforce languages; personal dosimetry and survey-meter checks before, during and after exposure; source accountability log reconciled every shift. PPE: dosimetry (monitoring, not protection). | NDT Supervisor / Radiation Protection Officer [role to be confirmed] Priority: High Target date: [SITE INPUT REQUIRED] | 1 x 4 = 4 — LOW | Radiography permit; controlled-area calculation; dosimetry records Review trigger / reference: Review for every radiography operation. Ref: approved ITP and radiation procedure [SITE INPUT REQUIRED]. | Evidence status: NOT CONFIRMED — verification-critical HOLD POINT: no exposure while any person is inside the controlled area or the survey meter is unavailable/defective. |

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| RA-35 | Use of chemical products: coatings, solvents, insulation and fireproofing materials — SDS-DEPENDENT | Coating/ insulation/ fireproofing locations [CONFIRM product list] | Flammable/toxic product exposure, skin/respiratory damage, fire from solvent vapours, or environmental release — specific hazards depend entirely on the actual products. | Occupational illness, fire, spill contamination. Persons at risk: Applicators, insulators and nearby workers; exposure via inhalation, skin contact and ignition sources. | NOT CONFIRMED: no product list or SDS file was available to this pre-assessment. KAEFER's historical Jafurah scope included coating, fireproofing and insulation [S-05]; current products unknown. | 3 x 3 = 9 — MODERATE | VERIFY FIRST: obtain the full product list and current SDS for every product before mobilisation. ENGINEERING: ventilated application/mixing areas, flammables storage in approved cabinets/containers away from ignition. ADMIN: product-specific risk assessment (COSHH-style) per SDS, crew briefing in their language, spill kits at use points, waste disposed via the site waste system. PPE: strictly per SDS (gloves, eye protection, respirator type as specified). | HSE Officer / Contractor Supervisor Priority: Moderate Target date: [SITE INPUT REQUIRED] | 2 x 3 = 6 — MODERATE | SDS file; product-specific assessments; hot-work controls near flammables Review trigger / reference: Review on every new product. Ref: product SDS [SITE INPUT REQUIRED before use]. | Evidence status: NOT CONFIRMED — verification-critical HOLD POINT: no product may be used on site before its SDS assessment is complete and briefed (applies also to RA-28 coatings). |
| RA-36 | Site traffic, mobile plant and desert driving (incl. journeys to remote fronts) | Site roads, laydown areas, access routes [CONFIRM traffic plan] | Vehicle–pedestrian collision, reversing incidents in blind spots, rollover or collision on desert/access roads aggravated by fatigue, dust and speed. | Fatality or serious injury to pedestrians, drivers and passengers. Persons at risk: Pedestrians in plant areas, drivers, passengers, other road users; exposure via shared roads and work areas. | TO BE VERIFIED: traffic management plan, driver authorisation and journey management not reviewed. | 4 x 5 = 20 — EXTREME | ENGINEERING: segregated pedestrian walkways and crossings; one-way systems and physical separation in laydown areas; reversing alarms/cameras and proximity systems where fitted. ADMIN: site traffic plan with speed limits set by the plan (values from the plan — not assumed), authorised-driver register with defensive-driving verification, banksman for reversing in congested areas, journey management with rest requirements for long desert drives, seat-belt rule enforced for all occupants, no night driving to remote fronts unless approved. PPE: high-visibility clothing in plant/traffic areas. | Construction Manager / Logistics Coordinator Priority: Critical Target date: [SITE INPUT REQUIRED] | 2 x 5 = 10 — HIGH | Traffic management plan; driver authorisation register; journey plans Review trigger / reference: Review monthly and after any traffic incident/near miss. Ref: site traffic plan [SITE INPUT REQUIRED]. | Evidence status: To be verified — no site evidence reviewed STOP-WORK: any vehicle defect affecting brakes/steering/alarms removes the vehicle from service; pedestrian entry into an active plant-movement zone stops the movement. |

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| RA-37 | Outdoor work in extreme heat (Saudi summer) — heat stress, dehydration, heatstroke | All outdoor workfronts | Heat exhaustion progressing to heatstroke during high-temperature work; risk peaks in summer months and during heavy physical work, fasting periods and for unacclimatised workers. | Heatstroke can be fatal or cause permanent organ damage; multiple casualties possible on peak days. Persons at risk: All outdoor workers, especially new arrivals, labourers, scaffolders and PPE-intensive trades; exposure via ambient heat plus metabolic load. | VERIFIED LEGAL CONTROL: Saudi HRSD's 2026 midday ban on work under direct sunlight ran 15 June–15 September 2026, 12:00–15:00, for private-sector establishments, and HRSD states heat-risk duties continue year-round [S-06]. Site heat-stress programme itself NOT evidenced — to be verified. | 4 x 4 = 16 — HIGH | ENGINEERING: shaded rest areas at each workfront, cooled drinking water and electrolytes freely available at the point of work. ADMIN: comply with the HRSD midday-work ban dates/hours in force for the actual work season (2027 notice to be verified if work falls in that season); acclimatisation schedule for new/returning workers; work/rest cycles driven by measured conditions (e.g. WBGT) under the site heat-stress procedure; buddy checks and supervisor training to recognise heat illness; emergency response for suspected heatstroke (rapid cooling and medical evacuation) rehearsed. PPE: breathable/light PPE selections where the task allows. | HSE Manager / Contractor Supervisors Priority: Critical Target date: [SITE INPUT REQUIRED] | 2 x 4 = 8 — MODERATE | Site heat-stress procedure; HRSD ban compliance schedule [current-season notice to be verified] Review trigger / reference: Review at season start and after any heat-illness case. Ref: HRSD midday-ban notice 2026 [S-06]; site heat-stress procedure [SITE INPUT REQUIRED]. | Evidence status: Legal duty verified; site programme to be verified STOP-WORK: work under direct sunlight stops during the legally banned hours in season; any suspected heatstroke case stops the crew's work and triggers medical response and a programme review. |
| RA-38 | Cutting, drilling and concrete work — respirable dust/silica; noise and hand-arm vibration across construction tasks | Civil, mechanical and fabrication areas [CONFIRM exposure survey] | Respirable crystalline silica from concrete/stone cutting and drilling; excessive noise from plant and tools; hand-arm vibration from breakers/grinders. | Silicosis and lung disease (chronic), noise-induced hearing loss, hand-arm vibration syndrome. Persons at risk: Operators and nearby workers; exposure via dust plumes, noise zones and tool vibration over repeated shifts. | TO BE VERIFIED: no exposure monitoring data exists in public sources — industrial-hygiene survey required before ratings can be confirmed. | 4 x 3 = 12 — HIGH | ENGINEERING: wet cutting/on-tool extraction for silica tasks; low-noise/low-vibration tools selected at procurement; noise barriers/distance for fixed plant. ADMIN: exposure assessment and monitoring programme by a competent hygienist; hearing-protection zones signed where monitoring requires; tool-use time limits and rotation for vibration; health surveillance for exposed groups. PPE: fit-tested respiratory protection for silica tasks and hearing protection in signed zones (protection selected from monitoring results). | HSE Manager / Occupational Hygienist [competent person] Priority: Moderate Target date: [SITE INPUT REQUIRED] | 2 x 3 = 6 — MODERATE | Industrial hygiene survey and monitoring plan Review trigger / reference: Review on monitoring results and task changes. Ref: site hygiene plan [SITE INPUT REQUIRED]. | Evidence status: To be verified — no site evidence reviewed STOP-WORK: dry cutting of concrete/stone is prohibited where wet/extraction controls are unavailable. |

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| RA ID | Work activity / step | Asset / area / workfront | Hazard / hazardous event | Potential consequence & persons at risk | Existing controls | Initial risk (L x S = score) | Additional controls / actions (hierarchy of controls) | Responsible role & target date | Residual risk (L x S = score) | PTW / procedure reference | Verification evidence / status |
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| RA-39 | Manual handling, housekeeping, slips, trips and falls on the level | All work areas, walkways and stores | Musculoskeletal injury from manual handling of valves, blinds, scaffold components; trips over cables/hoses/debris; slips on spilled liquids or sand-covered surfaces. | Strains, fractures, lost-time injuries; high-frequency/low-severity profile driving overall injury rates. Persons at risk: All trades; exposure via routine movement and handling throughout every shift. | TO BE VERIFIED: housekeeping inspection regime and mechanical-aid availability not evidenced. | 4 x 3 = 12 — HIGH | ENGINEERING: mechanical handling aids (trolleys, hoists, telehandlers) planned into tasks; cable/hose ramps and designated walkways kept clear. ADMIN: task-specific manual-handling assessment for heavy/awkward items (blinds, valves, bundles) with team-lift rules; housekeeping standard with daily area sign-off by supervisors; spill response immediate-clean rule. PPE: safety footwear and gloves per task. | Contractor Supervisors Priority: Moderate Target date: [SITE INPUT REQUIRED] | 3 x 2 = 6 — MODERATE | Manual-handling assessments; housekeeping inspection records Review trigger / reference: Review weekly via inspection trends. Ref: site HSE plan [SITE INPUT REQUIRED]. | Evidence status: To be verified — no site evidence reviewed STOP-WORK: walkways/escape routes found blocked are cleared before work continues in that area. |
| RA-40 | Night work, extended shifts, fatigue and multilingual workforce communication | All shifts and workfronts | Fatigue-related errors on safety-critical tasks; instructions misunderstood across languages/experience levels; inadequate lighting at night workfronts. | Serious incidents during low-supervision periods; delayed emergency response at night. Persons at risk: Night-shift and extended-shift crews, new starters, non-native speakers of the briefing language; exposure via reduced alertness and comprehension. | PARTIALLY EVIDENCED: a 2026 government report confirms an international contractor workforce active at the Jafurah Gas Plant project (Hyundai E&C E&I supervisor) [S-08]; site shift, lighting and briefing arrangements not reviewed. | 3 x 4 = 12 — HIGH | ENGINEERING: task lighting to the site standard at all night workfronts, verified by measurement where required. ADMIN: fatigue-management rules in the site plan (maximum hours, rest days, night-shift rotation) — values from the plan/owner standard, not assumed; competence and authorisation verified before safety-critical tasks regardless of shift; toolbox talks and permits briefed in the languages the crew actually understands, with comprehension checks; supervision ratios maintained at night for high-risk work or high-risk work restricted to day shift. PPE: task PPE. | Construction Manager / HSE Manager Priority: High Target date: [SITE INPUT REQUIRED] | 2 x 4 = 8 — MODERATE | Fatigue management plan; competence/authorisation records; lighting survey Review trigger / reference: Review monthly and after any night incident. Ref: PIA report [S-08] (workforce context); site plans [SITE INPUT REQUIRED]. | Evidence status: Partially evidenced — site arrangements to be verified HOLD POINT: critical lifts, line opening, confined-space entry and first energisation are day-shift-only unless a specific night-work risk assessment is approved by the Construction and HSE Managers. |

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|---|---|---|---|---|---|---|---|---|---|---|---|
| RA-41 | Emergency response interface: plant fire/gas emergency during construction occupancy | Whole defined site boundary | Construction workforce not aligned with the plant emergency plan — wrong muster point, missed headcount, re-entry into a hazardous area, or blocked emergency access during a plant alarm. | Multiple fatalities in a major fire/gas event; delayed rescue. Persons at risk: Every person inside the boundary, especially new and transient workers; exposure via the plant event itself and failed evacuation. | TO BE VERIFIED: the approved site emergency plan, alarm signals, muster points and headcount system were NOT available to this pre-assessment — all details are critical site inputs, none are assumed here. | 2 x 5 = 10 — HIGH | ADMIN: every worker inducted on the actual alarm signals, muster points and wind-direction rule from the approved plan before starting work; contractor headcount reconciled with the plant muster system (badge/gate data — see workforce register); joint emergency drills including construction crews at the frequency set by the plan; emergency access routes kept clear and audited weekly; wind socks maintained and visible at workfronts. ENGINEERING: alarm coverage audible/visible at all workfronts including high-noise areas (beacons). | HSE Manager / Plant Emergency Coordinator [role to be confirmed] Priority: Critical Target date: [SITE INPUT REQUIRED] | 1 x 5 = 5 — MODERATE | Approved site emergency response plan; drill records; induction records Review trigger / reference: Review after every drill, alarm event or workforce change. Ref: site emergency plan [SITE INPUT REQUIRED — critical]. | Evidence status: To be verified — no site evidence reviewed HOLD POINT: no workfront may operate outside alarm coverage or without a briefed muster route; a failed drill triggers corrective action before the next high-risk activity. |
| RA-42 | Hydrocarbon/ chemical handling, refuelling, waste management and spill response | Refuelling points, laydown, work areas and drainage paths [CONFIRM] | Spills of fuel, condensate, chemicals or contaminated water reaching soil or drains; incorrect waste segregation (incl. contaminated PPE/absorbents) creating fire and environmental harm. | Soil/groundwater contamination, fire, regulatory liability, harm to desert ecosystem. Persons at risk: Refuelling and maintenance crews; the environment; exposure via direct contact and contaminated ground. | TO BE VERIFIED: site environmental management plan, waste contractor arrangements and drain/seal status not reviewed. | 3 x 3 = 9 — MODERATE | ENGINEERING: bunded fuel/chemical storage sized to the site standard, refuelling on impermeable surfaces with drip trays, sealed/blanked drains in chemical areas where the design requires. ADMIN: spill kits at storage and refuelling points with trained responders; spill reporting and clean-up procedure in the environmental plan; waste segregation at source with labelled containers and licensed disposal route per Saudi requirements (requirements to be confirmed from the current regulations/permit). PPE: per SDS. | Environmental Officer / HSE Officer Priority: Moderate Target date: [SITE INPUT REQUIRED] | 2 x 3 = 6 — MODERATE | Site environmental management plan; waste transfer records Review trigger / reference: Review monthly and after any spill. Ref: site environmental plan [SITE INPUT REQUIRED]. | Evidence status: To be verified — no site evidence reviewed STOP-WORK: any spill reaching open soil or a drain stops the activity until contained and reported under the environmental plan. |

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| RA ID | Work activity / step | Asset / area / workfront | Hazard / hazardous event | Potential consequence & persons at risk | Existing controls | Initial risk (L x S = score) | Additional controls / actions (hierarchy of controls) | Responsible role & target date | Residual risk (L x S = score) | PTW / procedure reference | Verification evidence / status |
|---|---|---|---|---|---|---|---|---|---|---|---|
| RA-43 | Severe weather events: lightning, extreme wind, sandstorm, heavy rain/flash flooding in low areas | Whole site and access routes | Lightning strike on exposed workers/cranes; flash flooding of excavations, low roads and wadis after heavy rain; storm damage to temporary works and stored materials. | Fatalities, drowning/entrapment in flooded excavations, collapse of temporary works. Persons at risk: All outdoor personnel; drivers on access routes; exposure via open terrain and low-lying work areas. Heavy rain at Jafurah is recorded in KAEFER's project story [S-05]. | TO BE VERIFIED: site severe-weather procedure, warning sources and shelter locations not evidenced. | 2 x 5 = 10 — HIGH | ADMIN: weather monitoring with defined alert triggers and a named decision-maker per shift (triggers from the site procedure — not assumed); pre-storm securing checklist for cranes, scaffolds, temporary structures and loose materials; excavations and low areas evacuated and closed on a heavy-rain warning and inspected before re-entry; no driving through flooded sections — turn-around rule briefed to all drivers. ENGINEERING: lightning protection/shelter provision per the site plan at remote fronts. | Construction Manager Priority: High Target date: [SITE INPUT REQUIRED] | 1 x 5 = 5 — MODERATE | Site severe-weather procedure; pre-storm checklists Review trigger / reference: Review at season changes and after any weather event. Ref: KAEFER [S-05]; site procedure [SITE INPUT REQUIRED]. | Evidence status: To be verified — no site evidence reviewed STOP-WORK: all outdoor work stops on a lightning or flash-flood warning for the area; re-entry to excavations only after competent-person inspection (links to RA-24/RA-25). |

**NOTE ON RESIDUAL RATINGS: several major-accident rows (e.g. RA-03, RA-04, RA-10) retain HIGH residual scores because severity cannot be reduced — only likelihood can. Those risks are acceptable, if at all, only through the client's formal acceptance route with all stated controls verified in place. PPE alone never justifies a rating reduction in this register.**

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## 8. Critical controls / verification checklist

*Each critical barrier for a high-consequence event must be verified with evidence before the associated work proceeds. 'To be verified' means exactly that — the barrier is not credited until the evidence exists and a competent verifier has signed it.*

| # | Critical control / barrier | Applicability | Evidence required | Verifier (role) | Frequency | Verification status |
|---|---|---|---|---|---|---|
| 1 | Live-plant / SIMOPS interface review completed and work authorised by operations for the area and shift | To be verified | Signed SIMOPS review / permit coordination record for the area | Area Authority | Each shift / each new workfront | Not verified |
| 2 | Process isolation / positive isolation proven and LOTO applied before intrusive work | To be verified | Isolation certificate with zero-energy verification recorded | Authorised Isolator | Every intrusive task | Not verified |
| 3 | Line-opening controls: drain/vent/purge proven, first-break technique, gas test at break point | To be verified | Line-opening permit + gas test certificate | Mechanical Supervisor | Every line break | Not verified |
| 4 | Gas testing before and during work where the owner procedure requires it (LEL/O2/toxics scope per process data) | To be verified — scope depends on composition data (see RA-04) | Gas test certificates; tester authorisation records; monitor calibration records | Authorised Gas Tester | Per permit / continuous where required | Not verified |
| 5 | H2S monitoring and respiratory protection — ONLY if sour service is confirmed by process data | To be verified — composition data required | Composition data sheet; H2S plan; training and monitor records | Process Safety / HSE Manager | Before any intrusive work; continuous in sour areas | Not verified |
| 6 | Hot-work permit, ignition control, trained fire watch and post-work fire check in place | To be verified | Hot-work permit; fire-watch log; extinguisher checks | Hot Work Supervisor | Every hot-work task | Not verified |
| 7 | Confined-space controls: classification, isolation, atmospheric testing, attendant, communications, rescue plan with equipment at entry point | To be verified | Entry permit; test records; rescue drill/equipment check records | Entry Supervisor | Every entry | Not verified |
| 8 | Lifting controls: approved lift plan, certified crane/rigging, ground-bearing verification, barricaded drop zone, appointed person for complex/critical lifts | To be verified | Lift plan; certificates; set-up sign-off | Appointed Person (Lifting) | Every complex/critical lift; daily for routine | Not verified |
| 9 | Work-at-height controls: collective protection, inspected/tagged scaffolds, fall-arrest with verified anchors, rescue-from-height plan at the workfront | To be verified | Scaffold tags/register; harness inspection records; rescue plan | Scaffold Supervisor / HSE Officer | Daily checks; weekly scaffold inspection | Not verified |
| 10 | Excavation controls: permit, services clearance and scan, protective system, access/egress, daily competent-person inspection | To be verified | Excavation permit; scan records; inspection log | Competent Person (Excavations) | Daily and after rain | Not verified |
| 11 | Electrical controls: authorised persons only, prove-dead recorded, temporary-power inspection and tagging | To be verified | Electrical permits; prove-dead records; inspection register | Authorised Electrical Person | Every task; monthly installation inspection | Not verified |
| 12 | Pressure-test controls: approved test pack, calibrated gauges, barricaded exclusion zone, single command for pressurisation, controlled depressurisation | To be verified | Test pack; calibration certificates; boundary-clear record | Test Supervisor | Every test | Not verified |
| 13 | Radiography controls (if used): licensed crew, calculated barricaded controlled area, dosimetry, source accountability log | To be verified — applicability to be confirmed from ITP | Radiography permit; dosimetry and source logs | NDT Supervisor / RPO | Every exposure; source log each shift | Not verified |
| 14 | Emergency readiness: alarm coverage at all workfronts, briefed muster points, headcount reconciliation, joint drills, clear emergency access | To be verified — emergency plan is a critical site input | Induction and drill records; muster reconciliation reports | HSE Manager | Drills per plan; headcount every shift | Not verified |

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| # | Critical control / barrier | Applicability | Evidence required | Verifier (role) | Frequency | Verification status |
|---|---|---|---|---|---|---|
| 15 | Heat-stress controls: HRSD midday-ban compliance in season, water/shade/rest at workfronts, acclimatisation, WBGT-driven work/rest, heatstroke response | Applicable — legal duty verified [S-06]; site programme to be verified | Heat-stress procedure; WBGT logs; training records | HSE Manager | Daily in hot season | Not verified |
| 16 | Contractor competency: verified qualifications/authorisations, language-appropriate induction and toolbox talks, supervision in place | To be verified | Competence matrix; induction records; authorisation registers | Construction Manager / HSE Manager | At mobilisation; monthly audit | Not verified |
| 17 | Environmental containment: bunded storage, spill kits and trained responders, drain protection, waste segregation | To be verified | Inspection records; waste transfer notes; spill log | Environmental Officer | Weekly inspection | Not verified |
| 18 | Management of change / impairment control: no bypass or override of safety-critical systems without approved MOC/impairment permit and compensating measures | To be verified | Bypass/impairment register; MOC records | Process Safety Engineer | Reviewed every shift while active | Not verified |

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## 9. Task sequencing / JSA / permit matrix

*Permit names below are generic functional descriptions. The owner's current permit-to-work system governs the actual permit types, forms and approval chain — obtain the controlled site PTW procedure (Section 14) and re-map this table to it before use. A task-level JSA/JHA must be produced for each activity step by the executing contractor and briefed to the crew before work.*

| Task / activity | Prerequisite permit(s) | Isolation requirements | Drawings / documents | Competent persons | Pre-job briefing | Field verification | Return to service / handover |
|---|---|---|---|---|---|---|---|
| Work adjacent to live process equipment | Owner PTW; SIMOPS authorisation | As defined by operations for the task | Plot plan extract; method statement; this RA | Area Authority; Contractor Supervisor | Joint operations/contractor toolbox talk | Area Authority field check before start | Permit close-out with operations |
| Hot work in/near process areas | Hot-work permit; PTW | Equipment isolation as required by permit | Gas test certificate; fire-watch assignment | Hot Work Supervisor; Fire Watch; Gas Tester | Permit briefing at the work point | Gas test before and during work; fire-watch log | Post-work fire check signed; permit closed |
| Line opening / flange breaking | Line-opening permit; PTW | Positive isolation + drain/vent/purge proven | P&ID; line list; isolation certificate; blind register | Authorised Isolator; Mechanical Supervisor | First-break briefing incl. line-of-fire positions | Two-person line ID verification; gas test at break | Reinstatement checklist; blind register reconciled |
| Confined-space entry | Entry permit; PTW | Blinding/isolation of all connected lines; lockout of agitators etc. | Entry plan; rescue plan; gas test records | Entry Supervisor; Attendant; Gas Tester; Rescue Team | Entrant/attendant briefing and comms check | Continuous monitoring; entry log maintained | All persons/equipment accounted for; permit closed |
| Excavation near services | Excavation permit | Service isolation where required by owner | Services drawings; scan records; excavation design | Competent Person (Excavations) | Crew briefing on marked services | Daily inspection; hand-dig zone enforcement | Backfill/reinstatement record |
| Complex / critical lifting | Lifting permit (if required by site system) | Area cleared; plant isolation if over live equipment (operations decision) | Approved lift plan; crane/rigging certificates; ground assessment | Appointed Person; Crane Operator; Riggers; Signaller | Lift briefing with all crew incl. abort signals | Trial lift; drop-zone check; weather check | Lift completion report; defects logged |
| Electrical work / HV switching | Electrical permit / sanction for test | Isolation at all sources; earthing per rules | Single-line diagrams; isolation certificate | Authorised Electrical Person | Briefing on boundaries and backfeed sources | Prove-dead recorded at the work point | Restoration under permit; records updated |
| Pressure testing | Pressure-test permit | Test boundary isolation; blinds per test pack | Approved test pack; calibration certificates | Test Supervisor; Mechanical Engineer | Boundary briefing; exclusion rules | Boundary-clear confirmation before pressurisation | Depressurisation and drain record; test certificate |
| Radiography (if applicable) | Radiography permit | Area cleared of non-essential personnel | Controlled-area calculation; source log | NDT Supervisor; RPO [to be confirmed] | Crew and area briefing | Survey-meter verification; dosimetry issued | Source secured and log reconciled |
| Commissioning / energisation | Commissioning permit; energisation notice | Boundary isolation per boundary certificate | PSSR records; boundary certificates; punch list | Commissioning Lead; Operations Representative | Boundary meeting with all affected parties | Permit cross-check before energisation | Operations acceptance certificate |
| Work at height / scaffold use | WAH permit (if required by site system) | n/a | Scaffold tag/register; rescue plan | Scaffold Supervisor; users briefed | Toolbox talk incl. drop-zone rules | Tag verified before access; anchor checks | Scaffold handback after inspection |
| Nitrogen purging (if used) | Commissioning permit | Isolation of purge boundary | Purge plan; gas test records | Commissioning Lead; Gas Tester | Briefing on asphyxiation risk and exclusion zones | O2 monitoring at access points | Oxygen proven normal and recorded before handover |

## 10. Responsible-role matrix

*Accountability is assigned to ROLES, never to invented individuals. Role titles below are functional descriptions; the site organisation chart and the owner PTW procedure must confirm the actual titles, appointments and authority before this document is approved.*

| Role | Accountability in this assessment | Authority / confirmation note |
|---|---|---|
| Area Authority / Operations | Owns the plant area; authorises work and SIMOPS; first point of contact for | Role title and authority to be confirmed against the site organisation chart and |

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| Role | Accountability in this assessment | Authority / confirmation note |
|---|---|---|
| Representative | plant upsets | PTW procedure |
| Permit Issuer / Permit Receiver (as defined by the site PTW system) | Issue, acceptance, suspension and close-out of permits within their competence | Exact titles and authorisation route per the controlled site PTW procedure — not assumed |
| Construction Manager | Overall construction delivery, resources, sequencing and interface management | Approves high-risk work scheduling with HSE and operations |
| Contractor Supervisor (each contractor) | Direct supervision of their crews; toolbox talks; first-line control verification | One named supervisor per crew per shift — names from contractor mobilisation records |
| HSE Manager / HSE Officer | RA ownership, inspections, incident reporting, heat-stress and hygiene programmes, drill coordination | Stop-work authority to be confirmed in the site HSE plan |
| Process Safety Engineer | Major-accident hazard interface, MOC/impairment control, PSSR participation | Confirms where HAZID/HAZOP/LOPA studies are required |
| Mechanical Engineer / Supervisor | Piping, line opening, pressure testing and joint integrity | Approves test packs jointly with commissioning |
| Electrical / Instrumentation Engineer | Electrical safety, isolations, loop testing and energisation readiness | Authorised-person register maintained by this function |
| Lifting Supervisor / Appointed Person | Lift planning, categorisation, set-up verification and lift execution control | Appointed in writing per the owner lifting standard |
| Crane Operator / Riggers / Signaller | Execute lifts only to the approved plan; stop on any unsafe condition | Certification verified before mobilisation |
| Authorised Gas Tester | Atmospheric testing for permits, entries and hot work | Authorisation and calibration records current |
| Confined-Space Attendant / Rescue Team | Entry control, communications and rescue execution | Rescue team trained and equipped at the entry point |
| QA/QC Inspector | Hold-point inspection for joints, welds, coating and reinstatement | Releases hold points in the ITP only |
| Commissioning Lead | System completion, PSSR, energisation and handover boundaries | Single accountable authority at the construction/operations boundary |
| Emergency Response Coordinator [site title to be confirmed] | Alarm response, muster/headcount reconciliation, drill programme with plant operations | Interface defined in the approved site emergency plan |

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## 11. Action tracker

*No action is closed merely because it is written here. Closure requires the stated verification evidence and sign-off by the closure reviewer. Overdue Critical actions block the associated work (see hold points in Section 7).*

| Action ID | Linked RA | Action (specific) | Priority | Accountable role | Target | Verification evidence | Status | Closure reviewer |
|---|---|---|---|---|---|---|---|---|
| ACT-01 | RA-01/02 | Obtain approved plot plan and fix the assessment boundary, workfront list and asset IDs; re-issue this RA against them | Critical | Operations / Document Control | Before approval | Plot plan revision on file; RA Rev. 1 boundary table | Open | HSE Manager |
| ACT-02 | RA-03/04 | Obtain gas composition data (incl. H2S/benzene) and record the H2S-controls decision for every system in the package | Critical | Process Safety Engineer | Before any intrusive work | Composition data sheet; signed decision note | Open | HSE Manager |
| ACT-03 | All | Replace the provisional 5x5 matrix with the owner/client risk matrix and re-rate all 43 entries in a validation workshop | Critical | HSE Manager (owner) | Before approval | Workshop record; re-rated register | Open | Project Manager |
| ACT-04 | All | Obtain current authorized owner HSE/PTW/LOTO/hot-work/confined-space/line-opening/excavation/lifting/ pressure-test/MOC/emergency procedures and re-map Sections 8–9 to them | Critical | Document Control (owner) | Before approval | Controlled copies in document register | Open | HSE Manager |
| ACT-05 | RA-41 | Obtain the approved site emergency plan; extract alarm signals, muster points, routes, medical and rescue provisions into the induction pack | Critical | HSE Manager | Before work starts | Emergency plan on file; induction content updated | Open | Operations Representative |
| ACT-06 | RA-41 / Sec.5 | Start daily actual-headcount capture (gate/badge + contractor returns) and reconcile with the muster system; populate Section 5.2 | Critical | Contractor Admin / Security | Before work starts; then daily | Dated manpower reports; reconciliation log | Open | HSE Manager |
| ACT-07 | RA-09/10/11 | Verify isolation hardware, blind register practice and authorised-isolator appointments by field inspection | Critical | Area Authority | Before first line break | Inspection report; appointments in writing | Open | Mechanical Engineer |
| ACT-08 | RA-17/18 | Complete ground-bearing assessments and crane-pad verification for all planned critical-lift set-up positions | High | Appointed Person (Lifting) | Before first critical lift | Signed set-up verifications | Open | Construction Manager |
| ACT-09 | RA-37 | Confirm the heat-stress programme: current-season HRSD notice, WBGT monitoring, water/shade/rest provision, acclimatisation roster | Critical | HSE Manager | Before hot season / immediately for current work | Programme document; WBGT logs started | Open | Construction Manager |
| ACT-10 | RA-24/25 | Complete underground-services survey/scan for all excavation areas in the first work package | Critical | Civil Supervisor / Operations | Before excavation | Scan records; marked drawings | Open | Construction Manager |
| ACT-11 | RA-13 | Prove confined-space rescue capability (trained team + equipment) by a practical drill at the first entry location | Critical | Entry Supervisor | Before first entry | Drill report with timings and corrective actions | Open | HSE Manager |
| ACT-12 | RA-26/27 | Approve test packs (hydrostatic default; pneumatic only with engineered justification) and verify gauge calibration certificates | High | Mechanical / Commissioning Engineer | Before first test | Approved test packs; certificates | Open | Commissioning Lead |
| ACT-13 | RA-20/21 | Inspect and tag all existing scaffolds in the work area; close out red-tag items; verify anchor points for fall arrest | High | Scaffold Supervisor | Before work at height | Scaffold register updated | Open | HSE Officer |
| ACT-14 | RA-36 | Issue the site traffic plan (routes, walkways, speed limits, driver authorisations) and brief all drivers | High | Logistics Coordinator | Before peak construction | Plan issued; briefing records | Open | Construction Manager |
| ACT-15 | RA-38 | Commission an industrial-hygiene baseline survey (silica, noise, vibration, welding fume) for the package tasks | Moderate | HSE Manager | Within first month of work | Survey report; monitoring plan | Open | HSE Manager |
| ACT-16 | RA-35/28 | Collect the full product/SDS file for coatings, solvents, insulation and fireproofing products; complete product- | High | HSE Officer | Before product mobilisation | SDS file; assessment records | Open | Contractor Supervisor |

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| Action ID | Linked RA | Action (specific) | Priority | Accountable role | Target | Verification evidence | Status | Closure reviewer |
|---|---|---|---|---|---|---|---|---|
| | | specific assessments | | | | | | |
| ACT-17 | RA-07 | Audit the live bypass/impairment register for the interface areas; close or formally extend every overdue item | Critical | Process Safety Engineer | Before hot/intrusive work | Register audit report | Open | Operations Representative |
| ACT-18 | RA-40 | Verify night-work controls: lighting survey, fatigue rules and day-shift-only list for critical tasks | High | Construction Manager | Before night work | Lighting survey; shift rules issued | Open | HSE Manager |
| ACT-19 | RA-42 | Verify bunded storage, spill kits, drain protection and waste contracts for the work areas | Moderate | Environmental Officer | Before mobilisation completes | Inspection records; waste contracts | Open | HSE Officer |
| ACT-20 | All | Conduct the site walkdown and worker consultation for this assessment; record observations and amend the register | Critical | HSE Manager | Before approval | Walkdown report; consultation records; RA Rev. 1 | Open | Project Manager |

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## 12. Emergency response and rescue interface

**The approved site emergency response plan was NOT available to this pre-assessment. Nothing in the table below is assumed: every operational detail is a critical site input to be extracted from that plan and briefed at induction (links to RA-41 and ACT-05).**

| Emergency element | Detail required from the approved site emergency plan | Status |
|---|---|---|
| **Emergency number / control-room contact** | Exact number(s) and radio channels for raising the alarm from any workfront | SITE INPUT REQUIRED — critical |
| **Alarm signals** | Signal types and meanings for fire, gas release, evacuation and all-clear; audibility/visibility at high-noise workfronts | SITE INPUT REQUIRED — critical |
| **Muster points** | Locations per area, wind-direction alternates, and the route to each from every workfront | SITE INPUT REQUIRED — critical |
| **Headcount / muster reconciliation** | System used (badge/gate data), who reconciles contractor numbers, and the reporting line to the incident commander | SITE INPUT REQUIRED — critical |
| **Wind indication** | Wind-sock locations and the rule for selecting upwind muster routes | SITE INPUT REQUIRED — critical |
| **Medical provision** | Clinic location [CONFIRM], ambulance provision and route to the receiving hospital [CONFIRM] — no response times are stated until evidenced | SITE INPUT REQUIRED — critical |
| **Fire response** | Plant fire team interface, hydrant/firewater status at the interface, extinguisher provision for construction areas | SITE INPUT REQUIRED — critical |
| **Gas-release response** | Actions on gas-detector alarm, safe refuge locations and re-entry authority | SITE INPUT REQUIRED — critical |
| **Rescue — confined space** | Team, equipment and drill standard (see RA-13) | SITE INPUT REQUIRED — critical |
| **Rescue — work at height** | Equipment and trained rescuers per workfront (see RA-21) | SITE INPUT REQUIRED — critical |
| **Communications dead zones** | Radio coverage survey for remote fronts and the fallback check-in system (see RA-29) | SITE INPUT REQUIRED — critical |
| **Contractor–operator interface** | Who commands, who accounts for contractor personnel, and how construction permits are suspended on alarm | SITE INPUT REQUIRED — critical |

## 13. Environmental and occupational-health screening

*Screening only: no exposure level is stated anywhere in this document, because no monitoring data exists in public sources. Where the screening identifies a credible exposure, an industrial-hygiene assessment/monitoring action is raised (ACT-15).*

| Aspect | Source activity (register link) | Screening conclusion | Control / monitoring requirement |
|---|---|---|---|
| Dust / respirable silica | Concrete cutting, drilling, excavation (RA-24, RA-38) | Credible exposure — unquantified | Wet cutting/extraction; hygiene survey and monitoring; fit-tested RPE where required |
| Welding fume / metal fume | Welding and thermal cutting (RA-31) | Credible exposure — unquantified | LEV/extraction; consumable SDS assessment; monitoring in enclosed spaces |
| Noise | Plant, tools, generators (RA-38) | Credible exposure — unquantified | Noise survey; signed hearing-protection zones from results |
| Hand-arm vibration | Breakers, grinders, needle guns (RA-38) | Credible exposure — unquantified | Tool selection, exposure-time management, health surveillance |
| Heat stress | All outdoor work (RA-37) | HIGH — legally regulated [S-06] | HRSD ban compliance in season; WBGT-driven work/rest; acclimatisation; hydration/shade |
| Ergonomics / manual handling | Blinds, valves, scaffold materials (RA-39) | Frequent moderate risk | Mechanical aids; task assessments; team lifts |
| Hydrocarbon / chemical spills | Refuelling, tie-ins, product use (RA-42, RA-35) | Credible — consequence-driven | Bunding, spill kits, drain protection, reporting and clean-up per environmental plan |
| Waste generation | Construction waste, contaminated | Routine — controlled | Segregation at source; licensed disposal route per current Saudi requirements [CONFIRM] |

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| Aspect | Source activity (register link) | Screening conclusion | Control / monitoring requirement |
|---|---|---|---|
| | absorbents/PPE (RA-42) | | |
| Soil / drain contamination | Spills and washdown (RA-42) | Credible | Sealed work areas, no uncontrolled washdown, incident reporting |
| Water use and sanitation (remote desert site) | Welfare provision (Sec. 5/14) | Welfare-critical in heat | Potable water at workfronts, sanitation and rest facilities verified at mobilisation [CONFIRM provision] |

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# 14. Document limitations, assumptions and site-data request list

**LIMITATIONS — BLUNT STATEMENT. This document is a publicly researched pre-assessment. It is based solely on the public sources listed in Section 17 (accessed 28 October 2026) and on established HSE practice for operating gas facilities. It has NOT been verified by a site walkdown, operator or worker interview, approved drawings, current permits, process safety information, or a shift-specific headcount. It is not an official Aramco document, not Aramco-approved, not a contractor method statement, and not a substitute for the controlled project HSE system, for task-level JSAs, or for specialist process studies (HAZID/HAZOP/LOPA/QRA). Access-controlled Aramco construction/safety manuals were not available and were not reviewed [S-10]. Any person who treats this draft as authorization to work would be misusing it.**

ASSUMPTIONS MADE (all flagged for confirmation): (1) the selected boundary is the Jafurah Field Gas Plant plus adjacent expansion workfronts at the same location; (2) the activity list reflects the publicly announced Phase Two scope elements and standard brownfield interface tasks — each row applies only where the approved work pack confirms the activity; (3) the provisional 5x5 matrix in Section 6.1 is used because the owner matrix was unavailable; (4) permit and role names are generic functional descriptions pending mapping to the owner's controlled procedures.

## 14.1 Site-data request list

| # | Site input required | Why it matters | Owner role | Priority |
|---|---|---|---|---|
| 1 | Approved site plot plan, exact assessment boundary, workfront and asset/unit IDs | Defines what this RA actually covers; without it every area reference stays [CONFIRM] | Operations / Document Control | Critical — before approval |
| 2 | Current work package: task list, method statements, JSA/JHA set and construction look-ahead schedule | Confirms which register rows apply and which are N/A | Construction Manager | Critical — before approval |
| 3 | Current P&IDs, line list, equipment list; design pressure/temperature data | Required for isolation, line opening and pressure-test rows; no values are assumed in this draft | Process / Mechanical Engineer | Critical — before intrusive work |
| 4 | Gas composition data incl. H2S, benzene and other hazardous constituents; SDS file for all products | Decides RA-04/RA-05/RA-35 controls; verification-critical hazards cannot be closed without it | Process Safety Engineer | Critical — before intrusive work |
| 5 | Hazardous-area classification drawings and electrical single-line drawings | Governs hot-work locations, temporary power and electrical work controls | E&I Engineer | Critical — before hot/electrical work |
| 6 | Underground services drawings and survey/scan records | Prevents service strikes in excavation and crane set-up | Civil / Operations | Critical — before excavation |
| 7 | Owner/client risk matrix and risk-acceptance (ALARP) criteria | Replaces the provisional 5x5 matrix and bands used in this draft | HSE Manager (owner) | Critical — before approval |
| 8 | Current owner HSE manual and procedures: PTW, LOTO/isolation, hot work, confined space, line opening, excavation, lifting, pressure testing, radiography, commissioning, MOC, emergency plan | Some owner manuals are access-controlled for registered suppliers [S-10]; this draft does not claim to have reviewed them | Document Control (owner) | Critical — before approval |
| 9 | SIMOPS plan and live permit register access | Needed to operate RA-02 controls and daily permit coordination | Area Authority | Critical — before work starts |
| 10 | Actual headcount by shift: gate/access-control report, contractor mobilisation register, daily manpower report | Current total persons inside the boundary is not publicly disclosed; emergency headcount depends on it | Contractor Admin / Security | Critical — before work starts; then daily |
| 11 | Contractor/subcontractor register with scope, competence and authorisation records | Verifies who is on site and whether crews are authorised for their tasks | Construction Manager | High |
| 12 | Lifting studies, crane/rigging certificates and ground-bearing/geotechnical data | Closes RA-17/RA-18 hold points for the actual lifts planned | Appointed Person (Lifting) | High — before first critical lift |
| 13 | Gas detection and fire-protection status at the interface; impairment register | Confirms credited barriers for hot work and line opening (RA-07) | Operations / Fire & Gas | Critical — before hot/intrusive work |
| 14 | Approved emergency response plan: alarm signals, muster points, routes, medical/ambulance provision, rescue capability, communications | RA-41 cannot be closed without it; no locations or response times are assumed in this draft | HSE Manager / Emergency Coordinator | Critical — before work starts |
| 15 | Current-season heat-stress requirements: HRSD notice for | 2026 ban dates verified [S-06]; a 2027 notice must be | HSE Manager | High — before hot season |

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| # | Site input required | Why it matters | Owner role | Priority |
|---|---|---|---|---|
| | the actual work season + site heat-stress procedure and WBGT arrangements | checked if work falls in that season | | |
| 16 | Site walkdown report, worker consultation records, incident/near-miss history and inspection findings authorised for this assessment | Grounds the RA in observed conditions instead of public sources alone | HSE Manager | High — before approval |
| 17 | Existing specialist studies: HAZID/HAZOP reports, LOPA/QRA where they exist, PSSR procedure | Activity RA does not replace these; interfaces must be confirmed (RA-03) | Process Safety Engineer | High |
| 18 | Welfare and camp/transport arrangements: water, shade, rest areas, sanitation, medical clinic location [CONFIRM] | Supports heat-stress and welfare controls for a remote desert site | Construction Manager | Moderate |

# 15. Approval, validation and review

## 15.1 Implementation, validation & review

| Item | Requirement for this assessment |
|---|---|
| **Critical controls / hold points** | The critical controls in Section 8 and the hold point on each Section 7 register row must be verified with evidence before the associated work starts. No work within agreed protection zones of live equipment without signed operations authorisation. |
| **Pre-start verification** | PTW active, isolations proved (zero-energy verification), gas composition / H2S decision recorded (ACT-02), owner matrix adopted (ACT-03), emergency arrangements briefed, toolbox talk completed and actual headcount capture running (ACT-06). |
| **Review triggers** | See the review triggers listed in Section 15.3 — any trigger event requires re-validation of the affected register rows before work continues. |
| **Emergency and rescue arrangements** | All operational details are critical site inputs from the approved site emergency plan (Section 12). Nothing in this pre-assessment states a muster point, response time or alarm level. |

## 15.2 Consultation & approval

| Prepared by / role / date | Operations / supervisor review | HSE review | Authorized approver / date |
|---|---|---|---|
| Role: HSE — competent assessor<br>Name: _____<br>Signature: _____<br>Date: _____ | Role: Area Authority / Operations Representative (Construction Manager review also required — see note)<br>Name: _____<br>Signature: _____<br>Date: _____ | Role: HSE Manager<br>Name: _____<br>Signature: _____<br>Date: _____ | Role: as defined by the client's document system<br>Name: _____<br>Signature: _____<br>Date: _____ |

*Additional review functions to be completed in the client's document system: Reviewed by — Construction (Construction Manager); Reviewed by — Process / Mechanical / Electrical discipline engineers (as applicable). Approval block intentionally left blank: names, signatures and dates may only be entered by the real authorized persons after site validation. Signing this Rev. 0 pre-assessment without closing the Critical site inputs in Section 14.1 would not constitute a valid approval for execution.*

## 15.3 Review triggers

REVIEW TRIGGERS: change in scope, method, equipment or process status; new SIMOPS condition; any incident or significant near miss; change in gas composition or isolation boundary; new contractor or major workforce change; onset of the heat-ban season or a severe-weather event; expired or failed critical control; expired permit conditions; or the review interval set by the client's procedure — whichever occurs first.

# 16. Quality check

| QA check | Result |
|---|---|
| All numeric figures carry a source / date / scope label | PASS — every figure in Sections 3 and 5 is tied to a source register entry [S-01..S-09] |

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| QA check | Result |
|---|---|
| | |
| No corporate or sector headcount presented as current Jafurah headcount | PASS — Section 5.1 warning; Section 5.2 left as site-record form |
| No invented site count, workfront, process condition, chemical inventory, incident statistic, permit/drawing number, name, signature or approval | PASS — unknowns are [CONFIRM] / [SITE INPUT REQUIRED]; approval block blank |
| Jafurah vs Jubail / Riyas not confused | PASS — Riyas recorded as a separate Jubail site and excluded (Sections 3, 4) |
| Master Gas System Phase III not treated as one Jafurah location | PASS — recorded as a separate Kingdom-wide programme and excluded |
| Every High/Extreme hazard has controls, a responsible role, verification evidence and a hold point / acceptance requirement | PASS — checked row by row in Section 7 and mirrored in the XLSX register |
| Risk scores internally consistent with the stated matrix; no residual reduction without a credible control rationale | PASS — scores recomputed programmatically; severity is not reduced for major-accident rows |
| 'Existing control' and 'additional action' clearly distinguished | PASS — separate columns; existing controls carry evidence status |
| Legal requirements vs technical guidance not misclassified | PASS — HRSD decision labelled Saudi law; ISO 45001 labelled a management-system standard; OSHA labelled US technical guidance only |
| Source links work; claim-level citations present | PASS for sources opened on 28 October 2026 (verification status stated per source in Section 17); two dossier sources are honestly marked not re-verified |
| Tables legible; landscape used for wide registers; headers repeat; headers/footers and page numbers present | PASS — register, checklists, permit matrix and action tracker are landscape with repeated header rows; footer carries document ID, revision and page number |
| Document status states 'not approved for execution' | PASS — status banner on the cover and in every footer |
| Format follows the supplied generic reference templates without presenting them as a client form | PASS — reference cover/doc-control grid, proposed-matrix presentation, register column logic and approval block adopted; the templates are identified as generic layout references only, NOT an official Aramco/client form |

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# 17. References and traceability

*Access date for all sources: 28 October 2026. Verification status is stated honestly for each source: sources that could not be re-opened in this check are marked as cited in the research dossier, not silently dropped. Requirement types: [LAW] Saudi legal/regulatory, [CLIENT] owner-specific, [STANDARD] international standard, [GUIDANCE] technical guidance only.*

| Ref | Source (title — organisation, date) | Exact URL | Assertion(s) supported in this document | Verification status |
|---|---|---|---|---|
| S-01 | Utilizing unconventional energy resources in oil and gas — Jafurah section — Saudi Aramco, Page current at access date | https://www.aramco.com/en/what-we-do/operations/unconventional-resources | Jafurah production started December 2025; ambition by 2030: ~2.0 bscfd sales gas, 420 MMscfd ethane, ~630,000 bpd gas liquids and condensates; reserves 229 Tscf gas-in-place, 75 BSTB condensate-in-place (re-assessed 2023). Targets are plans, not current measured rates. | Verified — opened 28 Oct 2026 |
| S-02 | Aramco's strategic gas expansion progresses with $25bn contract awards — Saudi Aramco (news release), 30 June 2024 | https://www.aramco.com/en/news-media/news/2024/aramcos-strategic-gas-expansion-progresses-with-25bn-contract-awards | 16 contracts ~US$12.4bn for Jafurah Phase Two: gas compression facilities and associated pipelines; Jafurah Gas Plant expansion incl. gas processing trains; utilities, sulfur and export facilities; Riyas NGL fractionation facilities in Jubail. Master Gas System Phase III: ~US$8.8bn, ~4,000 km pipelines, 17 new gas compression trains (separate programme). | Verified — opened 28 Oct 2026 |
| S-03 | Aramco signs $11 billion Jafurah midstream deal with international consortium led by GIP — Saudi Aramco (news release), 14 August 2025 | https://www.aramco.com/en/news-media/news/2025/jafurah-midstream-deal | Jafurah Midstream Gas Company to lease development/usage rights for the Jafurah Field Gas Plant and Riyas NGL Fractionation Facility and lease them back to Aramco for 20 years; Aramco retains 51% and the exclusive right to receive, process and treat Jafurah raw gas. Day-to-day site control arrangements are NOT inferred from this announcement. | Verified — opened 28 Oct 2026 |
| S-04 | Aramco at a glance — Our operations — Saudi Aramco, Data as of 31 December 2025 | https://www.aramco.com/en/about-us/at-a-glance | Company employees 76,664 as of 31 December 2025. CORPORATE context only — not a Jafurah site headcount and excludes contractor personnel. | Verified — page content checked 28 Oct 2026 |
| S-05 | A river in the desert (Jafurah project story) — KAEFER SE & Co. KG, 2024 (project story) | https://kaefer.com/kaefer-stories/a-river-in-the-desert/ | KAEFER's own Phase 1 scope: access, coating, fireproofing, refractory and insulation; more than 400 of its own workers on site in 2024, with its team expected to rise to around 700 the following year (a plan, not a confirmed actual). Also records modular scaffold yard method (~1 km off-plot), suspended scaffolding, and weather challenges incl. heavy rain, wind and sandstorms. Historical, contractor-specific — NOT total site workforce and NOT confirmed current method. | Verified — opened 28 Oct 2026 |
| S-06 | Conclusion of the 2026 Ban on Work Under Direct Sunlight implementation period — Saudi Ministry of Human Resources and Social Development (HRSD) with NCOSH, 15 September 2026 | https://www.hrsd.gov.sa/en/media-center/news/%D8%A7%D9%86%D8%AA%D9%87%D8%A7%D8%A1%C2%A0%D9%81%D8%AA%D8%B1%D8%A9%C2%A0%D8%AA%D8%B7%D8%A8%D9%8A%D9%82%C2%A0%D9%82%D8%B1%D8%A7%D8%B1%C2%A0%D8%AD%D8%B8%D8%B1%C2%A0%D8%A7%D9%84%D8%B9%D9%85%D9%84-%D8%AA%D8%AD%D8%AA%C2%A0%D8%A3%D8%B4%D8%B9%D8%A9%C2%A0%D8%A7%D9%84%D8%B4%D9%85%D8%B3%C2%A0 | SAUDI LAW / REGULATORY DECISION: the 2026 ban on work under direct sunlight ran 15 June – 15 September 2026, 12:00–15:00, for private-sector establishments; HRSD stresses heat-stress prevention duties continue year-round. Verify the notice for the season in which work actually occurs. | Verified — opened 28 Oct 2026 |
| S-07 | ISO 45001:2018 — Occupational health and safety management systems — International Organization for Standardization (ISO), Published 2018-03; last reviewed and confirmed in 2024 | https://www.iso.org/standard/63787.html | ISO 45001:2018 remains current (confirmed 2024); a revision is in progress (stage 90.92 at access date) and Amendment 1:2024 (climate action changes) applies. It is a management-system standard — framework for this assessment's process, NOT a site hazard checklist or Saudi law. | Verified — opened 28 Oct 2026 |
| S-08 | Siquijor electrician credits TESDA certification for Saudi job — Philippine Information Agency (government news), 5 June 2026 (as dated in research dossier) | https://pia.gov.ph/features/siquijor-electrician-credits-tesda-certification-for-saudi-job/ | Identifies an electrical & instrument supervisor employed by Hyundai Engineering & Construction assigned to the Jafurah Gas Plant project — evidence of current contractor E&I activity at the project and of an international workforce. Does not evidence total headcount or the full contractor list. | Verified — opened 28 Oct 2026 |
| S-09 | Saudi construction-sector registered workers, Q2 2026 — DataSaudi / GASTAT-based data, Q2 2026 | https://datasaudi.sa/en | Figure cited in the research dossier: 3,348,143 registered construction-sector workers (Q2 2026). NATIONAL SECTOR context only — not Aramco's workforce and not Jafurah headcount. | NOT RE-VERIFIED in the 28 Oct 2026 check — as cited in research dossier; do not rely on it for site decisions |

| Ref | Source (title — organisation, date) | Exact URL | Assertion(s) supported in this document | Verification status |
|---|---|---|---|---|
| S-10 | Information and resources for our suppliers (incl. CWQAP resources) — Saudi Aramco, Page current at access date | https://www.aramco.com/en/what-we-do/suppliers/supplier-resources | Aramco's Contractor Workforce Qualification Assurance Program materials are password-protected for registered contractors — confirming that owner construction/safety manuals were NOT available to, or reviewed for, this public-source pre-assessment. | Verified — opened 28 Oct 2026 |
| S-11 | 2025 Sustainability Report (full, English) — Saudi Aramco, 2025 | https://www.aramco.com/-/media/publications/corporate-reports/sustainability-reports/report-2025/english/2025-saudi-aramco-sustainability-report-full-en.pdf | Corporate sustainability context as cited in the research dossier. No site-specific Jafurah safety data has been taken from it for this assessment. | NOT RE-VERIFIED — as cited in research dossier (URL as listed there; large PDF not opened in this check) |
| S-12 | Building at home (Saudi gas programme coverage) — EnterpriseAM KSA, Current at dossier date | https://enterpriseam.com/ksa/issues/building-at-home/ | Secondary reporting on Jafurah phases/programme status, as cited in the research dossier. Used only as corroboration; no load-bearing fact in this assessment rests on it alone. | NOT RE-VERIFIED — as cited in research dossier |
| S-13 | Process Safety Management — Overview — US OSHA, Current at access date | https://www.osha.gov/process-safety-management | TECHNICAL GUIDANCE ONLY (US regulator): PSM concepts used to structure major-accident interface controls (RA-01/03/07/08). NOT Saudi law and NOT a substitute for owner process-safety requirements. | Verified — opened 28 Oct 2026 |
| S-14 | Process Safety Management — Hazard Recognition — US OSHA, Current at dossier date | https://www.osha.gov/process-safety-management/hazards | TECHNICAL GUIDANCE ONLY: hazard-recognition concepts supporting RA-03/RA-04/RA-10. | As cited in research dossier (topic page not separately opened in this check) |
| S-15 | Confined Spaces — Overview — US OSHA, Current at dossier date | https://www.osha.gov/confined-spaces | TECHNICAL GUIDANCE ONLY: confined-space control concepts supporting RA-05/RA-13. | As cited in research dossier (topic page not separately opened in this check) |
| S-16 | Welding, Cutting and Brazing — Hazards and Solutions; Oil & Gas Hot Work guidance — US OSHA, Current at dossier date | https://www.osha.gov/welding-cutting-brazing/hazards-solutions/ | TECHNICAL GUIDANCE ONLY: hot-work and welding-fume control concepts supporting RA-30/RA-31. | As cited in research dossier (topic page not separately opened in this check) |

Requirement-type key for the register's technical references: the HRSD midday-ban decision [S-06] is Saudi LAW. Aramco owner procedures (not reviewed) would be CLIENT requirements. ISO 45001 [S-07] is a STANDARD used as a management framework. OSHA material [S-13..S-16] is GUIDANCE from a US regulator and is never presented as Saudi law. END OF DOCUMENT — Rev. 0 — Public-source pre-assessment.

**Template use note: The completed assessment must reflect the actual work scope and current site conditions. Risk ratings, legal/client references, workforce data, inspection observations and approvals must be verified by competent and authorized personnel. The generic reference format used for this document's layout does not establish compliance or client acceptance.**

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